Berk

Fundamentals of Corporate Finance, 2nd Edition {Jonathan Ber

- The matching principle: permanent vs temporary needs, cash buffers: Ch. 20, §20.2 (p. 598–601)

20.2 The Matching Principle

In a perfect capital market, the choice of financing is irrelevant; thus, how the firm chooses to finance its short-term cash needs cannot affect the firm's value. In reality, important market frictions exist, including transaction costs. For example, one transaction cost is the opportunity cost of holding cash in accounts that pay little or no interest. Firms also face high transaction costs if they need to negotiate a loan on short notice to cover a cash shortfall. Firms can increase their value by adopting a policy that minimizes these kinds of costs. One such policy is known as the *matching principle*. The **matching principle** states that short-term cash needs should be financed with short-term debt and long-term cash needs should be financed with long-term sources of funds.

matching principle States that a firm's short-term cash needs should be financed with short-term debt and long-term cash needs should be financed with long-term sources of funds.

permanent working capital The amount that a firm must keep invested in its short-term assets to support its continuing operations.

temporary working capital The difference between the firm's actual level of investment in short-term working capital needs and its permanent working capital requirements.

Permanent Working Capital

Permanent working capital is the amount that a firm must keep invested in its short-term assets to support its continuing operations. Because this investment in working capital is required so long as the firm remains in business, it constitutes a long-term investment. The matching principle indicates that the firm should finance this permanent investment in working capital with long-term sources of funds. Such sources have lower transaction costs than short-term sources of funds that have to be replaced more often.

Temporary Working Capital

Another portion of a firm's investment in its accounts receivable and inventory is temporary and results from seasonal fluctuations in the firm's business or unanticipated shocks. This **temporary working capital** is the difference between the firm's actual level of investment in short-term working capital needs and its permanent working capital investment. Because temporary working capital represents a short-term need, the firm should finance this portion of its investment with short-term financing.

Permanent Versus Temporary Working Capital

To illustrate the distinction between permanent and temporary working capital, we return to the Springfield Snowboards example. Table 20.2 presented cash flow forecasts assuming seasonal sales. In the spreadsheet in Table 20.6, we report the underlying levels of working capital that correspond to these forecasts.

In Table 20.6, we see that working capital for Springfield varies from a minimum of $2,125,000 in the first quarter of 2013 to $5,425,000 in the third quarter. The minimum level of working capital, or $2,125,000, can be thought of as the firm's permanent working capital. The difference between this minimum level and the higher levels in subsequent quarters (for example, $5,425,000 − $2,125,000 = $3,300,000 in the third quarter) reflects Springfield's temporary working capital requirements.

TABLE 20.6

Projected Levels of Working Capital for Springfield Snowboards, 2013, Assuming Seasonal Sales

	Quarter	2012Q4	2013Q1	2013Q2	2013Q3	2013Q4
2	**Net Working Capital Requirements ($000)**					
3	Minimum Cash Balance	500	500	500	500	500
4	Accounts Receivable	3,273	1,200	600	600	3,600
5	Inventory	300	950	2,900	4,850	300
6	Accounts Payable	−477	−525	−525	−525	−525
7	**Net Working Capital**	3,596	2,125	3,475	5,425	3,875

Financing Policy Choices

Following the matching principle should, in the long run, help minimize a firm's transaction costs.[3] But what if, instead of using the matching principle, a firm financed its permanent working capital needs with short-term debt? When the short-term debt comes due, the firm will have to negotiate a new loan. This new loan will involve additional transaction costs, and it will carry whatever market interest rate exists at the time. As a result, the firm is also exposed to interest rate risk.

aggressive financing policy Financing part or all of a firm's permanent working capital with short-term debt.

Aggressive Financing Policy. Financing part or all of the permanent working capital with short-term debt is known as an **aggressive financing policy**. An ultra-aggressive policy would involve financing even some of the plant, property, and equipment with short-term sources of funds.

When the yield curve is upward sloping, the interest rate on short-term debt is lower than the rate on long-term debt. In that case, short-term debt may appear cheaper than long-term debt. However, we know that with perfect capital markets, Modigliani and Miller's results from Chapter 16 apply: The benefit of the lower rate from short-term debt is offset by the risk that the firm will have to refinance the debt in the future at a higher rate. This risk is borne by the equity holders, and so the firm's equity cost of capital will rise to offset any benefit from the lower borrowing rate.

Why, then, might a firm choose an aggressive financing policy? Such a policy might be beneficial if the market imperfections mentioned in Chapter 16, such as agency costs and asymmetric information, are important. The value of short-term debt is less sensitive to the firm's credit quality than long-term debt; therefore, its value will be less affected by management's actions or information. As a result, short-term debt can have lower agency and lemons costs than long-term debt, and an aggressive financing policy can benefit shareholders. On the other hand, by relying on short-term debt the firm exposes itself to **funding risk**, which is the risk of incurring financial distress costs should the firm not be able to refinance its debt in a timely manner or at a reasonable rate. The box on the commercial paper and the financial crisis (page 604) highlights exactly this risk.

funding risk The risk of incurring financial distress costs should a firm not be able to refinance its debt in a timely manner or at a reasonable rate.

conservative financing policy When a firm finances its short-term needs with long-term debt.

Conservative Financing Policy. Alternatively, a firm could finance its short-term needs with long-term debt, a practice known as a **conservative financing policy**. For example, when following such a policy, a firm would use long-term sources of funds to finance its fixed assets, permanent working capital, and some of its seasonal needs. The firm would use short-term debt very sparingly to meet its peak seasonal needs. To implement such a policy effectively, there will necessarily be periods when excess cash is available—those periods when the firm requires little or no investment in temporary working capital. In an imperfect capital market, this cash will earn a below-market interest rate, thereby reducing the firm's value. It also increases the possibility that managers of the firm will use this excess cash nonproductively—for example, on perquisites for themselves.

In Figure 20.1, we illustrate aggressive and conservative financial policies for Springfield. With an aggressive policy, Springfield relies entirely on short-term financing to fund its inventory buildup, so its borrowing reaches a peak along with its inventory in the third quarter. With a conservative policy, it uses long-term borrowing to ensure that

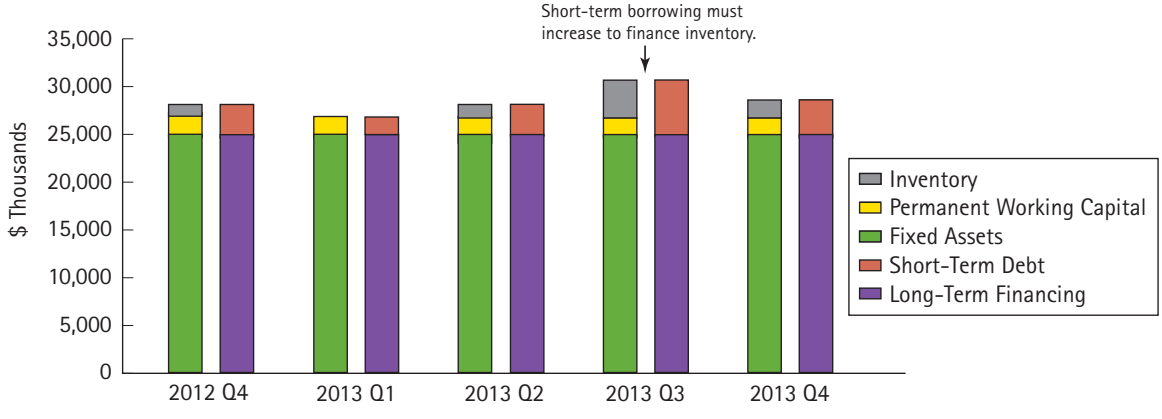
[3]Some evidence indicates that most firms appear to follow the matching principle. See W. Beranek, C. Cornwell, and S. Choi, "External Financing, Liquidity, and Capital Expenditures," *Journal of Financial Research* (Summer 1995): 207–222; and M. H. Stohs and D. C. Mauer, "The Determinants of Corporate Debt Maturity Structure," *Journal of Business* 69 (3) (1996): 279–312.

FIGURE 20.1 Financing Policy Choices for Springfield Snowboards

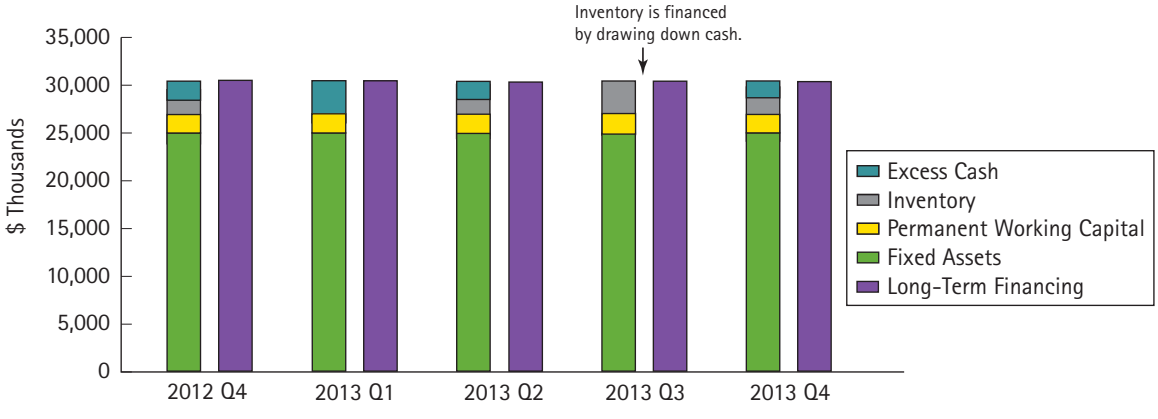
Assuming $25,000 in fixed assets, the figure shows the levels of fixed assets (green), permanent working capital ($2,125, yellow), and seasonal inventory (gray) for Springfield Snowboards. In panel (a), each quarter shows the mix of long-term financing (purple) and short-term debt (red) used to finance the assets and working capital. In the aggressive policy illustrated in panel (a), Springfield maintains no excess cash reserves and instead finances its working capital entirely through short-term borrowing. The amount of borrowing must match the amount of working capital.

Panel (b) illustrates a conservative policy, in which Springfield maintains enough cash reserves to cover its peak temporary financing needs. Springfield does this with $30,425 in long-term financing, which is enough to finance its fixed assets, permanent working capital, and peak inventory. Its total working capital is constant at $5,425, but the composition of the working capital varies over the year as it draws down its cash reserves (blue) to build inventory and then replenishes the cash reserves when it sells the inventory.

Panel (a): Aggressive financing policy: All working capital is financed with short-term debt.



Panel (b) Conservative financing policy: Working capital is financed with long-term debt.



it will have enough cash to pay for its peak inventory. In this case, Springfield's borrowing does not fluctuate at all, but rather it draws down its cash reserves so that its cash reserves are at a minimum when its inventory is at its peak.

Once a firm determines its short-term financing needs, it must choose which instruments it will use for this purpose. In the rest of this chapter, we survey the specific financing options that are available: bank loans, commercial paper, and secured financing.

Concept Check

3. What is the matching principle?
4. What is the difference between temporary and permanent working capital?

20.3 Short-Term Financing with Bank Loans

promissory note A written statement that indicates the amount of a loan, the date payment is due, and the interest rate.

One of the primary sources of short-term financing, especially for small businesses, is the commercial bank. Bank loans are typically initiated with a **promissory note**, which is a written statement that indicates the amount of the loan, the date payment is due, and the interest rate. In this section, we examine three types of bank loans: a single, end-of-period payment loan, a line of credit, and a bridge loan. In addition, we compare the interest rates of these bank loans and present the common stipulations and fees associated with them.

Single, End-of-Period Payment Loan

prime rate The rate banks charge their most creditworthy customers.

London Inter-Bank Offered Rate (LIBOR) The rate of interest at which banks borrow funds from each other in the London interbank market.

The most straightforward type of bank loan is a single, end-of-period-payment loan. Such a loan agreement requires that the firm pay interest on the loan and pay back the principal in one lump sum at the end of the loan. The interest rate may be fixed or variable. With a fixed interest rate, the specific rate that the commercial bank will charge is stipulated at the time the loan is made. With a variable interest rate, the terms of the loan may indicate that the rate will vary with some spread relative to a benchmark rate, such as the yield on one-year Treasury securities or the *prime rate*. The **prime rate** is the rate banks charge their most creditworthy customers. However, large corporations can often negotiate bank loans at an interest rate that is *below* the prime rate. For example, in its 2009 annual report, Hasbro indicated that the weighted average interest rate it paid on average short-term borrowings from domestic institutions was 1.2% in 2009. By comparison, the average prime rate in 2009 was 3.25%.[4] Another common benchmark rate is the **London Inter-Bank Offered Rate**, or LIBOR, which is the rate of interest at which banks borrow funds from each other in the London interbank market. It is quoted for maturities of one day to one year for ten major currencies. As it is a rate paid by banks with the highest credit quality, most firms will borrow at a rate that exceeds LIBOR.

Line of Credit

line of credit A bank loan arrangement in which a bank agrees to lend a firm any amount up to a stated maximum. This flexible agreement allows the firm to draw upon the line of credit whenever it chooses.

uncommitted line of credit A line of credit that is an informal agreement and does not legally bind a bank to provide the funds a borrower requests.

committed line of credit A legally binding written agreement that obligates a bank to provide funds to a firm (up to a stated credit limit) regardless of the financial condition of the firm (unless the firm is bankrupt) as long as the firm satisfies any restrictions in the agreement.

Another common type of bank loan arrangement is a **line of credit**, in which a bank agrees to lend a firm any amount up to a stated maximum. This flexible agreement allows the firm to draw upon the line of credit whenever it chooses.

Firms frequently use lines of credit to finance seasonal needs. An **uncommitted line of credit** is an informal agreement that does not legally bind the bank to provide the funds. As long as the borrower's financial condition remains good, the bank is happy to advance additional funds. A **committed line of credit** consists of a legally binding written agreement that obligates the bank to provide funds to a firm (up to a stated credit limit) regardless of the financial condition of the firm (unless the firm is bankrupt) as long as the firm satisfies any restrictions in the agreement. These arrangements are typically accompanied by a compensating balance requirement (that is, a requirement that the firm maintain a minimum level of deposits with the bank) and restrictions regarding the level of the firm's working capital. The firm pays a commitment fee of $\frac{1}{4}$% to $\frac{1}{2}$% of the unused portion of the line of credit in addition to interest on the amount that the firm borrowed. The line of credit agreement may also stipulate that at some point in time the

[4] Hasbro 2009 annual report and *Federal Reserve Statistical Release* Web site.

The Matching Principle

The following table includes quarterly working capital levels for your firm for the next year. Use it to answer Problems 3–7.

	Quarter			
($000)	1	2	3	4
Cash	$100	$100	$100	$100
Accounts Receivable	200	100	100	600
Inventory	200	500	900	50
Accounts Payable	100	100	100	100

3. What are the permanent working capital needs of your company? What are the temporary needs (*see MyFinanceLab for the data in Excel format*)?
4. If you chose to use only long-term financing, what total amount of borrowing would you need to have on a permanent basis? Forecast your excess cash levels under this scenario (*see MyFinanceLab for the data in Excel format*).
5. If you hold only $100 in cash at any time, what is your maximum short-term borrowing and when (*see MyFinanceLab for the data in Excel format*)?
6. If you choose to enter the year with $400 total in cash, what is your maximum short-term borrowing (*see MyFinanceLab for the data in Excel format*)?
7. If you want to limit your maximum short-term borrowing to $500, how much excess cash must you carry (*see MyFinanceLab for the data in Excel format*)?

Short-Term Financing with Bank Loans

8. The Hand-to-Mouth Company needs a $10,000 loan for the next 30 days. It is trying to decide which of three alternatives to use:

   Alternative A: Forgo the discount on its trade credit agreement that offers terms of 2/10, net 30.

   Alternative B: Borrow the money from Bank A, which has offered to lend the firm $10,000 for 30 days at an APR of 12%. The bank will require a (no-interest) compensating balance of 5% of the face value of the loan and will charge a $100 loan origination fee, which means Hand-to-Mouth must borrow even more than the $10,000.

   Alternative C: Borrow the money from Bank B, which has offered to lend the firm $10,000 for 30 days at an APR of 15%. The loan has a 1% loan origination fee.

   Which alternative is the cheapest source of financing for Hand-to-Mouth?

9. Consider two loans with one-year maturities and identical face values: an 8% loan with a 1% loan origination fee and an 8% loan with a 5% (no-interest) compensating balance requirement. Which loan would have the higher effective annual rate? Why?
10. Which of the following one-year, $1000 bank loans offers the lowest effective annual rate?

Grinblatt

Financial Markets And Corporate Strategy{Mark Grinblatt, She

- Why firms hedge: taxes, distress, debt capacity, capital needs, partial hedging: Ch. 21, §21.2 (p. 743–749)
- Which firms hedge: the empirical evidence: Ch. 21, §21.9 (p. 767–769)
- Factor-based hedging: Ch. 22, §22.8 (p. 805–807)

21.2 Why Do Firms Hedge?

Shortly after Iraq's invasion of Kuwait on August 2, 1990, the price of oil went up substantially. Within months of the invasion, jet fuel prices more than doubled, increasing Continental Airlines' fuel bill by $81 million a month. On December 3, 1990, Continental Airlines filed for Chapter 11 bankruptcy, citing rising fuel costs as the primary cause.[5]

If it were the rising fuel costs that bankrupted Continental, then the bankruptcy would have been avoided if Continental had hedged the risk associated with increased oil prices by making forward purchases of jet fuel prior to the Iraqi invasion. In retrospect, Continental's managers wished that they had hedged. However, the issue addressed in this section is whether or not Continental should have hedged knowing only what they knew prior to the Iraqi invasion.

A Simple Analogy

In our everyday life, individuals make choices that reduce risk. For example, when the authors of this text travel from UCLA to the Los Angeles Airport, they can choose between taking the San Diego Freeway or taking city streets. Most of the time they take the freeway because it is generally faster. However, travel time on the freeway varies considerably at rush hour, depending on traffic conditions. Traveling the freeway between 4:00 and 6:00 P.M. takes anywhere between 30 minutes and an hour, with an expected time of 40 minutes. With certainty, it takes 50 minutes to get to the airport on city streets.

Even though, on average, it takes longer to go by the city streets, they prefer this more certain route during rush hour because the loss from getting a bad outcome on the risky route (for example, missing their plane) far outweighs the gain of getting a good outcome on this route (for example, having enough time to get a drink before the flight leaves). Generally speaking, it is this type of asymmetry between gains and losses that leads us to make choices that reduce risk.

As we will show, similar asymmetries between losses and gains lead corporations to make choices that reduce their risks. This will be true even if the corporation is owned by stockholders who are risk neutral and thus prefer to maximize expected return. We describe the sources of these asymmetries below.

How Does Hedging Increase Expected Cash Flows?

The rest of this chapter will examine various ways that firms can increase their expected cash flows by implementing risk management programs. The primary benefits of hedging are related to taxes and to other market frictions discussed in Parts IV and V. Specifically, we will discuss the following benefits associated with hedging:

1. Hedging can decrease a firm's expected tax payments.
2. Hedging can reduce the costs of financial distress.
3. Hedging allows firms to better plan for their future capital needs and reduce their need to gain access to outside capital markets.
4. Hedging can be used to improve the design of management compensation contracts and it allows firms to evaluate their top executives more accurately.
5. Hedging can improve the quality of the investment and operating decisions.

[5] *The Wall Street Journal,* Dec. 4, 1990.

The gain from hedging in items (1) through (3) arises because the loss in the corporation's value from receiving one dollar less in profit is greater than the gain in value from one dollar more in profit. For example, the third motivation is based on the idea that the cost of not having enough internal capital available to fund a corporation's investment needs is greater than the benefits of having more than enough capital. Example 21.1 illustrates how this asymmetry between gains and losses motivates firms to hedge. Note that this example, as well as other examples in this chapter (unless specified otherwise), assume risk neutrality and a risk-free rate of zero. We use these assumptions not just for expositional simplicity, but to show that firms have incentives to reduce risk even when investors have no aversion to risk.

Example 21.1: How Hedging Creates Value

United Shoes exports U.S.-made shoes to the United Kingdom. Because of this, the firm's value is greater when the U.S. dollar is less valuable relative to the British pound. Suppose that it is equally likely that the British pound will be worth US$1.40, US$1.50, or US$1.60 next year. Under these scenarios, United Shoes is worth US$105 million, US$140 million, and US$160 million, respectively. By purchasing British pounds in the forward market at the current forward price of US$1.52 per British pound, the firm will realize for certain a value of US$138 million next year. Should United Shoes enter into the forward contract if its investors are risk neutral?

Answer: The US$138 million realized by hedging exceeds the US$135 million the firm would realize, on average, by not hedging. In this case, the firm benefits by hedging.

In Example 21.1, United Shoes was willing to take an unfair bet, paying $1.52 for British pounds that would be worth only $1.50, on average, to reduce uncertainty. United Shoes was willing to take this bet because movements in exchange rates that lower profits hurt the firm more than the firm is helped by equal-sized exchange rate changes in the opposite direction help it; that is, exchange rate uncertainty decreases United Shoes' expected value. Thus, the firm could improve its expected value by reducing this uncertainty.

In general, whenever a firm is hurt more by a negative realization of an economic variable (for example, an exchange rate change) than it is helped by a positive realization, the firm can increase its value by hedging. The following sections describe various situations where we might expect the costs of negative realizations to exceed the benefits of positive realizations. Each situation provides a motivation for why firms hedge.

How Hedging Reduces Taxes

Taxes play a key role in most financial decisions, and hedging is no different. Tax gains often accrue from hedging because of an asymmetry between the tax treatment of gains and losses. A U.S. corporation that has earned $100 million will pay about $35 million in federal income taxes. However, if that same corporation loses $100 million, the IRS will rebate its share of the losses only up to the amount of taxes the firm paid in the prior two years.[6] Hence, the firm often loses more value from a $100 million pretax loss than it gains in value from a $100 million pretax gain. Example 21.2 illustrates how firms can gain from hedging risks in situations of this kind.[7]

[6]Tax losses also can be carried forward, up to twenty years, but the present value of the future tax savings is less than the value of receiving the tax rebate immediately.

[7]Smith and Stulz (1985) discuss how hedging can be used by a corporation to reduce its expected tax liabilities.

Example 21.2: Taxes and Hedging

Cogen Pharmaceuticals sells a large fraction of its arthritis drugs in France for which it receives payments in Euros. Given that its costs are denominated in U.S. dollars, the firm's taxable earnings are subject to currency risk. Currency fluctuations are the firm's only source of risk, so the firm's pretax hedged and unhedged positions in two equally likely exchange rate scenarios can be described as follows:

Pretax Income for Two Equally Likely Scenarios (in $millions)

	Weak Dollar	*Strong Dollar*	*Average*
Unhedged	$100	−$20	$40
Hedged	35	35	35

The firm will thus achieve higher average pretax profits if it chooses not to hedge. Assume, however, that there is a 40 percent profits tax, but no tax deduction on losses. Show that the expected after-tax profits will be higher if the firm chooses to hedge:

Answer:

After-Tax Income for Two Equally Likely Scenarios (in $millions)

	Weak Dollar	*Strong Dollar*	*Average*
Unhedged	$60	−$20	$20
Hedged	21	21	21

Result 21.3 Because of asymmetric treatment of gains and losses, firms may reduce their expected tax liabilities by hedging.

Hedging to Avoid Financial Distress Costs

Chapters 16 and 17 examined why financial distress can be costly. Distress costs include costs arising from conflicts between debt holders and equity holders and those arising from the reluctance of many of the firm's most important stakeholders (for example, customers and suppliers) to do business with a firm having financial difficulties. By hedging its risks, a firm can increase its value by reducing its probability of facing financial distress in the future.

An Example Based on the Stakeholder Theory of Financial Distress. Consider, for example, Microtronics, a medium-sized U.S.-based manufacturer of scientific equipment, which needs to borrow $100 million to refinance an existing loan. Its operating value next year is assumed to depend on two factors: the health of the U.S. economy and the dollar/yen exchange rate. Microtronics' value is highest when (1) the U.S. economy is strong, since the demand for scientific equipment will then be strong, and (2) when the yen is strong relative to the dollar, which increases the dollar costs of Microtronics' Japanese competitors.

EXHIBIT 21.1 Microtronics' Unhedged Value in Four Scenarios*

	Strong U.S. Economy	*Weak U.S. Economy*
Weak yen	$150 million	$ 95 million
Strong yen	200 million	125 million

*Assumes customer confidence is high.

To evaluate Microtronics' loan application, the lender's analysts have calculated the firm's operating values under four scenarios. The analysts have assumed that the firm is unhedged and that customers maintain their confidence in the firm. These operating values are described in Exhibit 21.1.

If the U.S. economy is weak and the yen is weak, the firm's value ($95 million) will be less than its debt obligation ($100 million). The bank is particularly concerned about the weak economy because the $95 million value of the firm in this scenario assumes that customers maintain their confidence in the firm. As Chapter 17 noted, consumers of scientific equipment are reluctant to purchase from a firm with financial difficulties because of potential difficulties in obtaining spare parts and service. Thus, if Microtronics is unable to meet its debt obligations, its value in this scenario will be considerably less than the $95 million it would be worth if the firm were solvent. If the firm loses its customer base because of its financial distress, the bank may find it difficult to recover even a small fraction of what it is owed on the loan.

By hedging some of its currency risk (that is, by selling forward or futures contracts on the yen), Microtronics is betting that the yen will weaken relative to the dollar. Such a bet increases the firm's value in scenarios in which the yen is weak relative to the dollar and the firm is at a competitive disadvantage relative to its Japanese competitors. Hedging currency risk, in effect, transfers value from the scenarios where the yen is strong to scenarios where the yen is weak. If Microtronics hedges in a way that transfers $15 million from the strong-yen scenario to the weak-yen scenario, its values in the four scenarios would be as shown in Exhibit 21.2.

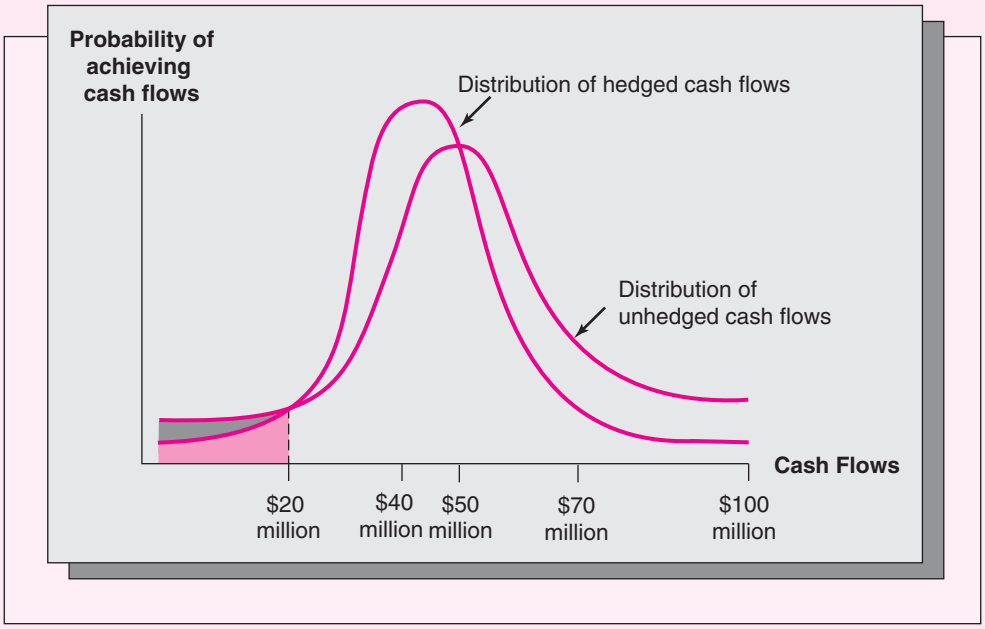
If Microtronics is hedged in this way, the bank is assured of being paid back in all four scenarios and the firm will not be exposed to the costs of bankruptcy. Hence, in the weak-yen, weak-economy scenario, hedging has eliminated the discrepancy between the actual value of the firm and the value of the firm when customer confidence is maintained.

Exhibit 21.3, which plots the distribution of profits with and without hedging, illustrates the potential advantages of hedging. The firm illustrated in this exhibit has a debt obligation of $20 million and will suffer financial distress costs if it cannot meet this obligation. The probability of not meeting the obligation is seen as the areas, to the left of $20 million, under the two curves. Since this illustration assumes that hedging is

EXHIBIT 21.2 Microtronics' Hedged Value in Four Scenarios

	Strong U.S. Economy	*Weak U.S. Economy*
Weak yen	$165 million	$110 million
Strong yen	185 million	110 million

EXHIBIT 21.3 Distribution of Profits with and without Hedging



costly, the mean of the unhedged distribution is greater than the mean of the hedged distribution. However, since the unhedged distribution has a larger variance, the area to the left of $20 million is greater than the corresponding area for the hedged distribution. In other words, hedging reduces the probability of financial distress.

Hedging does not always reduce the probability of financial distress. If the cost of hedging is sufficiently large, and if hedging reduces variance very little, then hedging may actually increase the probability of financial distress. Instances where this would occur are very unusual. However, if it is costly, then hedging may not be worthwhile for firms with very low financial distress costs.

Result 21.4 Firms that are subject to high financial distress costs have greater incentives to hedge.

Hedging to Increase the Tax Shield from Debt Capacity. The Microtronics example suggests that hedging reduces the expected costs of financial distress for any given debt level. As hedging opportunities improve, however, firms can choose more highly leveraged capital structures and take advantage of the tax and other advantages of debt financing (see Chapters 14 to 19). To understand why hedging allows a firm to take on more debt financing, consider again our analogy about the choice between taking the freeway or taking city streets to the airport. Taking the freeway requires travelers to allow at least an hour to get to the airport to be certain of catching the flight. However, when the slower but more certain option of taking city streets is available, travelers can leave 10 minutes later even though, on average, this route takes 10 minutes longer. Similarly, a firm that hedges its risks will be able to take on more debt while keeping its probability of financial distress within a reasonable level.

Hedging to Help Firms Plan for Their Capital Needs

Internal Financing, Underinvestment, and Overinvestment. Previous chapters explained why corporations view internal sources of equity capital as cheaper than external sources. Recall that the lower costs of internal funds arise for information reasons as well as because of taxes and transaction costs. Empirical evidence suggests that because of the difference between the costs of internal and external capital, investment expenditures by firms correspond closely to their cash flows. Firms thus have a tendency to overinvest or underinvest, depending on the availability of internally generated cash flows. The reliance on internally generated cash can be especially costly to firms that need to plan their investments in advance but have highly variable cash flows. To the extent that hedging reduces this variability in cash flows, it can increase the value of the firm.[8]

Consistent with this view, Lewent and Kearney (1990) noted in their explanation of Merck's strategy of actively hedging foreign exchange risk that a key factor in deciding whether to hedge is the "potential effect of cash flow volatility on our [Merck's] ability to execute our strategic plan—particularly, to make the investments in R&D that furnish the basis for future growth." It is surprising that a firm like Merck, which has easy access to debt markets and can borrow at one of the lowest corporate borrowing rates available, would express such concerns. However, if Merck expresses concerns about funding its strategic plans, firms with weaker credit ratings must have even greater concerns.

Suppose, for example, that Compaq is planning to invest $100 million in manufacturing equipment next year, and that it can finance the investment from internally generated cash flows if the dollar remains weak relative to the Euro. However, if the dollar strengthens, Compaq is likely to lose money on its European sales and will thus have considerably less cash available for investment. The executives at Compaq believe that outside investors will become overly pessimistic about the firm's future prospects if the firm shows a loss next year because of currency fluctuations. In this case the firm may not be able to issue equity or borrow at attractive terms and might be forced to pass up this positive net present value project because of the negative *NPV* of its outside financing alternatives. By hedging its currency risk, Compaq assures the availability of sufficient capital to fund this investment opportunity.

Result 21.5 Firms that find it costly to delay or alter their investment plans and that have limited access to outside financial markets will benefit from hedging.

The Desirability of Partial Hedging. If either the investment requirements or the costs and benefits of obtaining external financing are determined by the risk factor the firm wishes to hedge, a firm may not want to completely hedge the risk. For example, an oil firm may want to increase its exploration budget when oil prices are high. If the firm can raise capital on more favorable terms after having shown increased earnings, it will want to maintain some exposure to the risks of oil price fluctuations, so that its earnings look good when there is a need for outside capital. Such a firm would not choose to completely eliminate its exposure to oil price movements, but it still would like to hedge risks to guarantee sufficient investment funds in the event of substantial oil price declines.

[8]This is discussed in more detail in Lessard (1991) and Froot, Scharfstein, and Stein (1993).

Example 21.3: National Petroleum's Partial Hedge

National Petroleum currently has 3 million barrels of oil in reserves that will be extracted within the next three years at the rate of 1 million barrels of oil per year. Extraction costs are $6 per barrel and are expected to remain stable for the next three years. The current price of oil is $22 per barrel. Given current conditions, National is no longer exploring for new oil, but the company plans to reassess the situation after three years. If oil prices exceed $30 per barrel, National will reestablish its exploration operations, which will require a capital investment of $35 million.

Forward contracts for the delivery of oil in one, two, and three years are available at a price of $24 per barrel. National has overhead costs of about $3 million per year and interest obligations of $8 million per year. The firm will suffer significant financial distress-related costs if it cannot meet these fixed obligations should oil prices fall significantly. How much should National hedge?

Answer: National would like to hedge its oil price risk in a way that guarantees it will generate at least $11 million from its operations, so that it can cover its overhead costs and interest obligations. National can accomplish this by selling its entire supply of oil on the forward markets, which would guarantee cash flows of $18 million in each year. However, if the firm is completely hedged, it will not accumulate enough internally generated capital to fund new exploration efforts if oil prices exceed $30 per barrel. For this reason, National should hedge only enough to avoid financial distress, so that it has more money available if oil prices rise. Using the forward markets to sell 17/24 of its extracted oil annually is one way to achieve this.

As Example 21.3 illustrates, the fact that oil companies have greater investment needs when oil prices are higher provides one explanation for why these companies generally hedge very little of their oil price risk. As we discuss in Chapter 22, this also provides a motivation for the use of options to hedge oil price risk since they can be used to guarantee sufficient profits to avoid financial distress while allowing the firm to benefit from very high oil prices. The incentive of oil firms to hedge may also be reduced by the fact that the forward price of oil has generally been less than, rather than greater than, the spot price of oil, and as a result, there has historically been a positive expected cost of hedging (see footnote 3 for further discussion).

How Hedging Improves Executive Compensation Contracts and Performance Evaluation

Chapter 18 discussed conflicts of interest between shareholders and management and ways to design executive compensation contracts that minimize the costs of these conflicts. Recall that executive compensation should be designed to expose managers to the risk associated with factors they control (for example, success at cutting costs) while minimizing exposure to risks they do not control (for example, changes in interest rates). This suggests that a well-designed compensation package will not leave a risk-averse executive exposed to the risks of currency fluctuations, interest rate changes, and other factors over which the executive has no control.

Consider, for example, an Irish electronics company which sells most of its products in the United States. Its costs are mainly in Irish pounds, but its revenues are predominately in U.S. dollars. If the dollar strengthens against the pound (that is, if the company receives more pounds for every dollar it receives), then the company's profits increase since its revenues improve while its costs (in pounds) remain constant. Obviously, shareholders benefit from this, but it makes no sense to reward the company's top executives for this unexpected increase in profits because the exchange rate

mentioned above, if the real exchange rate remains constant, then a nominal exchange rate change is likely to have only a minor effect on IBM's cash flows. However, changes in real exchange rates can have a significant effect on these cash flows.

Consider what happens when the U.S. dollar strengthens against the Euro, making the computers more expensive in Euros. If the Euro weakened because of general inflation in Europe, so that the real exchange rate remained constant, then the price of computers in Europe, relative to other prices, would not have changed. In this case, demand for IBM computers would not be affected by the change in exchange rates. Contrast this case with one in which the real exchange rate does change, raising the relative price of IBM computers in Europe and lowering the demand for them. IBM's cash flows in Europe (calculated in U.S. dollars) would probably decrease in this case since it would either sell fewer computers at the same U.S. dollar price or, alternatively, be forced by competitors to cut its U.S. dollar price for computers.

As these arguments suggest, one of the major difficulties in assessing the effect of exchange rate changes on cash flows has to do with predicting the cause of the exchange rate movement. If we cannot predict whether future exchange rate fluctuations are associated with relative price changes, then forward and futures contracts provide imperfect hedges. This point is further illustrated in Example 21.7.

Example 21.7: Hedging Real Changes in the Yen

Suppose that American Lumber sells a significant quantity of prefabricated housing units in Japan. These units sell for ¥10,000 per square foot, with the market price increasing at the Japanese inflation rate. The exchange rate is currently ¥100 per US$. Analysts predict that it will trade in the range of ¥90 per US$ to ¥110 per US$ over the next 12 months, depending on the differences in the Japanese and U.S. inflation rates as well as productivity changes that can be reflected in trade imbalances. Forward prices are also at ¥100 per US$. Is it possible for American Lumber to create a hedge to guarantee a price of US$100 per square foot for the prefab units?

Answer: It is not possible to create such a hedge. To illustrate why it may not be possible to perfectly hedge, suppose that Japan experiences 5 percent deflation, causing housing unit prices to fall to ¥9,500. Although this would normally cause the yen to depreciate, suppose that a simultaneous increase in Japanese productivity, which tends to strengthen the yen, offset the effect of inflation on exchange rates exactly, so that the exchange rate stayed at ¥100 per US$. In this case, American Lumber will sell prefab units at $95 per square foot and will break even on its hedging activities regardless of its forward positions.

Result 21.14 When exchange rate changes can be generated by both real and nominal changes, it may be impossible for firms to effectively hedge their long-term economic exposures.

When it is difficult to hedge in the derivatives markets, firms sometimes undertake what is known as operational hedging, which involves changing the structure of the firm's operations. [See Chowdhry and Howe (1999) for details.]

21.9 Which Firms Hedge? The Empirical Evidence

A number of empirical studies have compared the characteristics of firms that use derivatives to firms that do not. Although research on this topic is still evolving, a number of patterns are worth considering.

Larger Firms Are More Likely to Use Derivatives Than Smaller Firms

A number of studies have found that larger firms are more likely to use derivatives than smaller firms.[16] The fact that smaller firms are less likely to use derivatives than larger firms is inconsistent with the view that smaller firms generally face higher risks of bankruptcy and thus have more to gain from hedging. However, the fixed costs of setting up a hedging operation and their lower level of sophistication probably explains why smaller firms are less likely to hedge. Indeed, Dolde (1993) found that among firms that have implemented hedging operations, the larger firms tend to hedge less completely than the smaller firms, leaving themselves more exposed to interest rate and currency risks. In other words, size is a barrier to setting up a hedging operation, but among firms that do hedge, smaller firms facing greater risks of bankruptcy hedge more completely.

Firms with More Growth Opportunities Are More Likely to Use Derivatives

Nance, Smith, and Smithson (1993) and Geczy, Minton, and Schrand (1997) provided evidence that firms with greater growth opportunities are more likely to use derivatives. In particular, firms with higher R&D expenditures and higher market-to-book ratios are more likely to use derivatives than companies that spend less on R&D, have lower market-to-book ratios, and, therefore, probably have fewer investment opportunities. This evidence is consistent with the idea that firms hedge to ensure that they have enough cash to fund their investment opportunities internally.

A number of other reasons explain why R&D-intensive firms with high market-to-book ratios are more likely to use derivatives. As Chapter 17 discussed, firms with these characteristics generally have higher financial distress costs, suggesting that they should hedge to ensure that they will meet their debt obligations. Furthermore, because R&D expenditures are tax deductible, these firms are likely to have lower taxable earnings, implying that the asymmetric tax treatment of gains and losses, a hedging motivation discussed earlier in this chapter, applies more to firms with high R&D expenditures.

Highly Levered Firms Are More Likely to Use Derivatives

Nance, Smith, and Smithson (1993); Block and Gallagher (1986); and Wall and Pringle (1989) found weak evidence that firms with more leveraged capital structures hedge more. The positive relation between leverage ratios and the tendency to hedge is consistent with the view that firms hedge to avoid financial distress costs. However, the weakness of the evidence probably reflects the tendency of firms with high financial distress costs, which have the most to gain from hedging, to have the lowest leverage ratios. For example, as Chapter 17 discussed, high R&D firms tend to use little debt and also tend to hedge because of their potential costs of financial distress.

Geczy, Minton, and Schrand (1997) found no significant relation between the debt ratios of most firms and their tendency to use derivatives. However, among those firms with high R&D expenditures and high market-to-book ratios, firms with more leverage are more likely to hedge. This implies that firms that suffer the highest costs of financial distress are more likely to hedge when they are highly leveraged.

[16] See Bodnar, Hayt, and Marston (1998); Nance, Smith, and Smithson (1993); Dolde (1993); and Geczy, Minton, and Schrand (1997).

Risk Management Practices in the Gold Mining Industry

The studies described above examined hedging choices across a number of different industries. A study by Tufano (1996) looked in greater detail at the risk management practices within a single industry, gold mining. Within a single industry, proxies for financial distress costs, financing constraints, and investment opportunities will probably vary much less than they do across industries. Consequently, differences in the hedging strategies across firms within a single industry are likely to be related to differences in the incentives and tastes of the top executives.

The evidence described in the Tufano study indicates that management incentives and tastes do have an important effect on the risk management practices in the gold mining industry. Specifically, managers who hold large amounts of their firm's stock tend to use forward and futures contracts to hedge more of their firm's gold price risk. Thus, managers who are personally the most exposed to gold price risk choose to hedge more of the risk. However, those who own relatively more stock options tend to hedge less, which may reflect the greater value of the options when volatility is increased. Tufano also found that firms with CFOs hired more recently hedge a greater portion of their exposure than firms with CFOs who have been on the job longer.

Risk Management Practices in the Oil and Gas Industry

A study by Haushaulter (2000) examined hedging choices of firms in the oil and gas industry. His evidence suggests that most oil and gas producers hedge only a small percentage of their future production. Specifically, only a quarter of the firms in his sample hedge more than 28 percent of their production. He also found that larger firms and firms that are more highly leveraged tend to hedge more, which is consistent with the studies mentioned earlier. Moreover, the evidence suggests that those firms whose production is located in regions where prices are highly correlated with the prices of exchange-traded futures contracts hedge more, which makes sense since these firms can probably hedge more effectively. However, in contrast to Tufano's study of the gold industry, Haushalter does not find a strong relation between the shareholdings and compensation of a firm's managers and the firm's risk management practices.

21.10 Summary and Conclusions

While we believe, as a general guideline, that most firms can benefit by hedging, the gains from hedging differ across firms. Firms can gain from hedging that reduces their probability of being financially distressed, especially in those industries where financial distress costs are the highest. There are also tax reasons for distressed firms to hedge, and gains that come from the fact that a firm's reported cash flows and profits are more informative when the firm has hedged out extraneous risks.

This chapter also considered situations where firms should not hedge. When excellent investment opportunities tend to arise at times when existing assets (unhedged) are most profitable, they may be better off remaining unhedged or only partially hedged. For example, an oil firm is likely to find more high-*NPV* exploration projects when oil prices and hence unhedged profits are the highest. An oil firm would not want to completely hedge away the variability in profits because that would leave it short of funds when favorable investment opportunities exist.

In addition, firms may choose not to hedge those risks about which they have private information or which they partially control. In this respect, firm are similar to individuals buying auto insurance. The safest drivers choose to be underinsured since they regard insurance as overpriced. Recognizing this tendency, insurance companies raise the rates for drivers wanting full insurance. In most cases, these considerations do not affect whether firms hedge in derivatives markets because firms are unlikely to have important private information about currency and commodity price movements. However, these considerations do affect whether firms insure against firm-specific risk or choose liability streams that leave them exposed to changes in their

own credit ratings. For the same reason that the safest drivers often choose to be underinsured, managers who believe that their firms are less risky than their credit rating reflects will choose to borrow short term in hopes that their credit rating will improve in the future. In other words, the safest firms will be overexposed to the risks connected with changes in their credit rating.

Our discussion of both foreign exchange and liability risk management indicated that implementing a sound risk management strategy requires a good understanding of the relations between interest rate changes, exchange rate changes, and inflation. The appropriate interest rate hedging strategy depends on the extent to which interest rate volatility is due to changes in the rate of inflation. Similarly, the appropriate foreign exchange hedging strategy depends on the extent to which currency fluctuations are due to differences between domestic and foreign inflation rates.

This chapter presented our view of how firms can use risk management tools to maximize firm value, which may differ from current practice. This difference is partly due to the limited experience many managers have in dealing with derivatives markets and risk management problems and partly due to potential incentive problems (see Chapter 18). For example, if managers get a large share of their compensation from stock options—to solve one kind of incentive problem—they may choose to speculate rather than hedge, since option values increase with risk.

Again, we must stress that risk management, like all corporate finance decisions, cannot be viewed in isolation. Corporations must view their risk management choices as part of an overall strategy that includes their choice of capital structure and executive compensation as well as considerations of overall product market strategy.

Key Concepts

Result 21.1: If hedging choices do not affect cash flows from real assets, then, in the absence of taxes and transaction costs, hedging decisions do not affect firm values.

Result 21.2: Hedging is unlikely to improve a firm's value if it does no more than reduce the variance of its future cash flows. To improve a firm's value, hedging also must increase expected cash flows.

Result 21.3: Because of asymmetric treatment of gains and losses, firms may reduce their expected tax liabilities by hedging.

Result 21.4: Firms that are subject to high financial distress costs have greater incentives to hedge.

Result 21.5: Firms that find it costly to delay or alter their investment plans and that have limited access to outside financial markets will benefit from hedging.

Result 21.6: The gains from hedging are greater when it is more difficult to evaluate and monitor management.

Result 21.7: Firms have an incentive to insure or hedge risks that insurance companies and markets can better assess. Doing this improves decision making. Firms will absorb internally those risks over which they have the comparative advantage in evaluating.

Result 21.8: If a firm's main motivation for hedging is to better assess the quality of management, the firm will probably want to hedge its earnings or cash flows rather than its value. However, if the firm is hedging to avoid the costs of financial distress, it should implement a hedging strategy that takes into account both the variance of its value and the variance of its cash flows.

Result 21.9: Corporations should organize their hedging in a way that reflects why they are hedging. Most hedging motivations suggest that hedging should be carried out at the corporate level. However, the improvement in management incentives that can be realized with a risk management program are best achieved when the individual divisions are responsible for hedging.

Result 21.10: Managers have private information only in exceptional cases. Given this, they almost always should be hedging rather than speculating.

Result 21.11: A firm's liability stream can be decomposed into two components: one that reflects default-free interest rates and one that reflects the firm's credit rating. When a firm borrows at a fixed rate, both components are fixed. When it rolls over short-term instruments, the liability streams fluctuate with both kinds of risks.

Derivative instruments allow firms to separate these two sources of risk: to

create liability streams that are sensitive to interest rates but not their credit ratings, as described in equation (21.4), and to create liability streams that are sensitive to their credit ratings but not interest rates, as described in equation (21.5).

Result 21.12: If changes in interest rates mainly reflect changes in the rate of inflation and if a firm's unlevered cash flows (and its EBIT) generally increase with the rate of inflation, then the firm will want its liabilities to be exposed to interest rate risk. If, however, interest rate changes are not primarily due to changes in inflation (that is, real interest rates change) and if the firm's unlevered cash flows are largely affected by the level of real interest rates, then the firm will want to minimize the exposure of its liabilities to interest rate changes.

Result 21.13: Exchange rate movements can be decomposed into those caused by differences in the inflation rates in the home country and the foreign country, and those caused by changes in real exchange rates. In most cases, the incentive is to hedge against real exchange rate changes rather than the component of exchange rate changes that is driven by inflation differences between the two countries.

Result 21.14: When exchange rate changes can be generated by both real and nominal changes, it may be impossible for firms to effectively hedge their long-term economic exposures.

Key Terms

economic risk 763
hedge 739
liability management 758
liability stream 758
nominal exchange rate 763
real exchange rate 763
risk management 739
risk profile 739
transaction risk 761
translation risk 762

Exercises

21.1. Small firms currently hedge less than large firms. Why is this? Do you expect smaller firms to start hedging more in the future? Explain.

21.2. Why is it harder to hedge currency risks in countries with volatile inflation rates?

21.3. Whistler Resorts is a Canadian ski resort just north of Vancouver, British Columbia. Discuss the resort's exposure to exchange rate risk.

21.4. It is now much easier to hedge risks than it was in the past. How should this affect a firm's optimal capital structure? Why?

21.5. The XYZ corporation manufactures in both Turkey and Japan for export to the United States. Japan has a stable monetary policy and, as a result, its inflation is easy to predict. Monetary policy in Turkey is much less predictable. In which of the two countries can XYZ more easily hedge against the risk that manufacturing costs, measured in U.S. dollars, will become significantly more expensive? Why?

21.6. Purchasing power parity (PPP) implies that real exchange rates remain constant. If PPP holds, do firms need to hedge their long-term foreign exchange exposure? Explain.

21.7. Oil firms hedge only part of their exposure to oil price movements. Why might that be a good idea?

21.8. Harwood Outboard manufactures outboard motors for relatively inexpensive motor boats. The firm is optimistic about its long-term outlook, but its bond rating is only BB. Describe how you would manage Harwood's liability stream if you believed that within two years Harwood's credit rating would improve to A.

22.8 Factor-Based Hedging

This section discusses how the factor betas of commitments and financial instruments can be used for risk management.

Computing Factor Betas for Cash Flow Combinations

One of the most useful things about factor models is the additivity property of factor betas. To compute the aggregated factor sensitivities of combinations of cash flows, add their respective factor betas. Example 22.14 illustrates the computation.

Example 22.14: Computing Factor Loadings for Combinations of Cash Flows

Consider the following two-factor models for the cash flows of projects a, b, and c, which are part of the ABC corporation. (Intercepts and coefficients are in millions of dollars.)

$$\begin{aligned}\tilde{C}_a &= 3 + \tilde{F}_{curr} - 4\tilde{F}_{int} + \tilde{\epsilon}_a\\ \tilde{C}_b &= 5 + 3\tilde{F}_{curr} + 2\tilde{F}_{int} + \tilde{\epsilon}_b\\ \tilde{C}_c &= 10 + 1.5\tilde{F}_{curr} + 0\tilde{F}_{int} + \tilde{\epsilon}_c\end{aligned}$$

Write out the factor equations for (1) the combination of all three projects and (2) a super-project that involves doubling the size of project a and combining it with project b alone.

Answers:

(1) $\alpha_p = 3 + 5 + 10 = 18$
$\beta_{p,curr} = 1 + 3 + 1.5 = 5.5$
$\beta_{p,int} = -4 + 2 = -2$

So $\tilde{C}_p$, the cash flow of the collection of projects, satisfies the factor equation

$$\tilde{C}_p = 18 + 5.5\tilde{F}_{curr} - 2\tilde{F}_{int} + \tilde{\epsilon}_p,$$

where $\tilde{\epsilon}_p$ is the sum of the three ϵ's.

(2) $\alpha_p = 2(3) + 5 = 11$
$\beta_{p,curr} = 2(1) + 3 = 5$
$\beta_{p,int} = 2(-4) + 2 = -6$

Thus

$$\tilde{C}_p = 11 + 5\tilde{F}_{curr} - 6\tilde{F}_{int} + \tilde{\epsilon}_p,$$

where

$$\tilde{\epsilon}_p = 2\tilde{\epsilon}_a + \tilde{\epsilon}_b$$

The additivity property of factor models makes it easy to compute the risk impact of adding any financial instrument to a firm's cash flow. When combining the cash flow from a real asset with the cash flow from a financial instrument, add their respective betas together to obtain the impact on the risk of the firm.

Computing Hedge Ratios

The last subsection noted that the sum of the cash flow factor betas, factor by factor, yields the cash flow betas of the aggregated cash flows. This means that the quantities of various financial instruments which perfectly hedge the factor risk exposures have

exactly the opposite sensitivity to the factors that the firm has. Therefore, the computation of such quantities is generally a straightforward mathematical calculation, as Example 22.15 illustrates.

Example 22.15: Eliminating Factor Loadings with Portfolios of Factor Portfolios

Consider the following two-factor model, which describes, per dollar of investment, how the cash flows of the projects of the Family School Alliance Corp. (FSA) relate to the currency and interest rate factors

$$\tilde{C}_{\text{FSA}} = .3 + \tilde{F}_{\text{curr}} - 4\tilde{F}_{\text{int}} + \tilde{\epsilon}_{\text{FSA}}$$

Assume that pure factor portfolios can be constructed (see Chapter 6) that perfectly track the factors. How many dollars need to be invested in the pure factor portfolio for currency (*curr*) and the pure factor portfolio for interest rates (*int*) to perfectly hedge the factor risk of FSA?

Answer: Per dollar of real investment, if FSA sells short $1 of the pure factor portfolio for currency risk and buys $4 of the pure factor portfolio for interest rates, it will eliminate all factor sensitivity.

Result 22.7 The factor risk of a cash flow is eliminated by acquiring a portfolio of financial instruments with factor betas exactly the opposite of the cash flow factor betas.

Direct Hedge Ratio Computations: Solving Systems of Equations

As Chapter 6 noted, pure factor portfolios often must be constructed from portfolios of more basic financial instruments. In this case, factor hedging can be implemented using the more basic financial instruments, as Example 22.16 illustrates.

Example 22.16: Eliminating Factor Loadings with Portfolios of Financial Instruments

Consider a two-factor model, where the two factors are interest rate movements and changes in inflation. Assume that the Noah's Bagel Company has a future cash flow with factor betas of 2.5 on the interest rate factor and 4.5 on the inflation factor. Noah's would like to eliminate its sensitivity to both factors by acquiring financial securities, yet it does not wish to use its own cash to do this.

Noah's contacts its investment bank and learns that it can (1) enter into a five-year interest rate swap contract, (2) purchase 30-year government bonds, and (3) acquire a sizable chunk of shares in DESPERATE, Inc. Investment 1, the swap contract, has no up-front cost and, per contract, has a factor equation for its future value described by

$$\tilde{C}_1 = 5 - 5\tilde{F}_{\text{inflation}} - 3\tilde{F}_{\text{inflation}}$$

Investment 2, the 30-year government bond, per million dollars invested, has a factor equation for its future cash flow of

$$\tilde{C}_2 = 10 - 5\tilde{F}_{\text{inflation}} - 1\tilde{F}_{\text{inflation}}$$

Investment 3 is the stock of DESPERATE, Inc., which, per million dollars invested, has a factor equation for its end-of-period value of

$$\tilde{C}_3 = 0 + 1\tilde{F}_{\text{inflation}} + 1\tilde{F}_{\text{inflation}}$$

Design a proper hedge against inflation and interest rate movements in this environment.

Answer: To design a future cash flow that is insensitive to inflation or interest rate movements, we need to find a costless portfolio of financial investments with an interest rate sensitivity of -2.5 and an inflation sensitivity of -4.5. The three investments are denoted by

x_1 = the number of contracts in the costless swap investment
x_2 = millions of dollars in the 30-year government bonds
x_3 = millions of dollars in DESPERATE, Inc.

The portfolio of these three investments has a cost of $0x_1 + 1{,}000{,}000x_2 + 1{,}000{,}000x_3$. Its sensitivity to the interest rate factor is

$$-5x_1 - 5x_2 + 1x_3$$

Its sensitivity to the inflation factor is

$$-3x_1 - 1x_2 + 1x_3.$$

Therefore, the hedge portfolio is found by simultaneously solving

$$\begin{aligned} 0x_1 + 1{,}000{,}000x_2 + 1{,}000{,}000x_3 &= 0 \\ -5x_1 - 5x_2 + 1x_3 &= -2.5 \\ -3x_1 - 1x_2 + 1x_3 &= -4.5 \end{aligned}$$

Since the first equation says $x_3 = -x_2$, one can substitute for x_3 in the other two equations, implying

$$-5x_1 - 6x_2 = -2.5$$

and

$$-3x_1 - 2x_2 = -4.5$$

Multiplying the first equation (of the two immediately above) by $-.6$ and adding it to the second equation yields

$$1.6x_2 = -3$$

or

$$x_2 = -1.875$$

Plugging this back into either of the above equations yields $x_1 = 2.75$. Thus, the solution is

1. Buy 2.75 swap contracts, representing $2.75 million in notional amount
2. Short $1.875 million in 30-year government bonds
3. Buy $1.875 million of DESPERATE, Inc.

22.9 Hedging with Regression

Regression provides a shortcut method for estimating a hedge ratio that minimizes risk exposure. By regressing the cash flow that one is trying to hedge against the value of the financial instrument used in the hedge, one derives a beta coefficient that determines the quantity of the hedging instrument that minimizes variance.[14]

[14]Chapter 4 notes the conditions required for variance minimization and for the portfolio variance formulas used here.

	US$	*€*
Disney	6.0%	9.7%
Metallgesellschaft	8.4	10.0

Design a currency swap agreement that will benefit both firms and also yield a 0.4 percent profit for the bank acting as an intermediary for the swap.

22.12. Consider a two-factor model, where the factors are interest rate movements and changes in the exchange rate. Your company has a future cash flow with factor betas of 2 on the interest rate factor and 5 on the exchange rate factor. You would like to eliminate your sensitivity to both of these factors by means of financial securities, but do not wish to use any of the company's cash to do this.

The following investment opportunities are available to you.

- You can purchase 30-year government bonds.
- You can enter into a 2-year interest rate swap agreement.
- You can invest in a foreign index fund.

The following factor equations, with the two factors being changes in interest rates and changes in exchange rates, correspond to the future values of the three investment opportunities, respectively.

$$\tilde{C}_1 = 4 - 4\tilde{F}_{\text{int}} \quad \text{(per \$1 million invested)}$$
$$\tilde{C}_2 = 6 - 2\tilde{F}_{\text{int}} + 6\tilde{F}_{\text{ex}} \quad \text{(per \$1 million invested)}$$
$$\tilde{C}_3 = 3 - 3\tilde{F}_{\text{int}} - 2\tilde{F}_{\text{ex}} \quad \text{(per \$1 million invested)}$$

Design a proper hedge against interest rate movements and exchange rates in this environment.

22.13. Ford is considering building a factory to produce its new Taurus. The factory will cost $100 million and will produce 10,000 automobiles one year from now, but its cost and production can be scaled up or down. As an analyst at Ford, you run a regression of the historical sales margin of 10,000 cars against the return of Ford's stock. You find that the regression coefficient is –$130 million, that the sales margin of a Taurus one year from now has a standard deviation of $1,000, and that the volatility of Ford's stock is 0.5. The risk-free rate over the coming year will be 10 percent. You can assume that the return on the factory is the total sales margin from all cars produced by the factory divided by the cost of the factory, less one.

Your task is the following:

a. To identify the minimum variance portfolio combination of Ford stock and the factory, and to interpret the results.

b. To identify the tangency portfolio mix of factory and stock if the expected selling margin of the Taurus is $22,000 and the expected return of Ford stock is 30 percent over the next year.

22.14. The National Basketball Association (NBA) has hired you as a consultant to figure out what the proper mix of Eastern and Western teams should be. An Eastern team has a return variance of .04, a Western team has a return variance of .09, and the correlation between the returns of an Eastern and a Western team is .25. Assume that all Eastern teams are identical and all Western teams are identical. What should the ratio of Eastern to Western teams be if the NBA wants to minimize return variance?

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Ross

ROSS, WESTERFIELD, JORDAN_ LAMB - Fundamentos de Administraç

- Motivos para manter caixa e custos de manter caixa: Cap. 19, §19.1 (p. 662–663)

vestimentos maiores podem ser pulverizados em vários CDBs de até $70.000,00. Isso é uma consideração importante para pequenas e médias empresas.

19.1 Motivos para manter saldos de caixa

John Maynard Keynes, em seu clássico trabalho *A teoria geral do emprego, juros e moeda*, identificou três motivos para a preferência por liquidez: o motivo especulação, o motivo precaução e o motivo transação. Discutiremos esses motivos a seguir.

Os motivos especulação e precaução

motivo especulação
a necessidade de manter reservas de liquidez para aproveitar oportunidades adicionais de investimento, como compras vantajosas.

O **motivo especulação** é a necessidade de manter saldos de caixa para poder aproveitar, por exemplo, as compras vantajosas que apareçam, as taxas de juros atraentes e (no caso das empresas com atuação internacional) as flutuações favoráveis da taxa de câmbio.

Para a maioria das empresas, a capacidade de tomar emprestado e de emitir títulos de dívida (como uma emissão de debêntures, por exemplo) pode ser um meio para atender ao motivo especulação. Assim, pode haver um motivo especulativo para manter a liquidez, mas não necessariamente para reter caixa propriamente dito, na forma de disponibilidades. Pense nisso desta maneira: se você tiver um cartão de crédito com um limite de crédito muito grande, provavelmente aproveitará todas as barganhas incomuns que aparecerem sem ter que carregar dinheiro.

motivo precaução
a necessidade de manter uma reserva de liquidez como segurança.

Isso também vale, em menor grau, para o motivo precaução. O **motivo precaução** é a necessidade de manter uma reserva financeira como segurança. Mais uma vez, provavelmente há motivos para manter liquidez por precaução. Entretanto, dado o fato de que o valor dos investimentos no mercado de renda fixa é relativamente certo e de que instrumentos, como os títulos do Tesouro, têm alta liquidez, não há necessidade de reter quantias substanciais de caixa disponível para fins de precaução.

O motivo transação

motivo transação
a necessidade de manter reservas de liquidez para atender às atividades normais de desembolso e financiamento do prazo de cobrança associadas às operações de funcionamento de uma empresa.

O **motivo transação** está associado ao caixa necessário para pagar as contas. As necessidades relacionadas às transações vêm das atividades normais de desembolso e de financiamento do prazo de cobrança da empresa. O desembolso de caixa inclui o pagamento de salários, dívidas comerciais, pagamento de empréstimos e juros, impostos e dividendos.

O caixa é recebido das vendas de produtos, da venda de ativos e de novas dívidas. Os fluxos de entrada de caixa (recebimentos) e os fluxos de saída de caixa (desembolsos) não estão sincronizados perfeitamente, e algum nível de caixa é necessário para servir de reserva.

Mesmo com transferências eletrônicas de fundos e outros mecanismos de pagamento "sem papel" e de alta velocidade que continuam sendo desenvolvidos, a demanda de caixa para transações não desaparecerá. Mesmo que desapareça, porém, ainda haverá uma demanda por liquidez e a necessidade de administrá-la com eficiência.

Saldos médios

Os saldos médios são outro motivo para reter saldos de caixa. Como discutimos no capítulo anterior, saldos de caixa podem ser mantidos nos bancos comerciais para remunerar parte dos serviços bancários que a empresa recebe. Uma exigência de saldo médio mínimo pelos bancos pode impor um limite mais baixo para o nível de caixa que uma empresa efetivamente dispõe.

Custos de manter caixa

Quando uma empresa mantém caixa além do mínimo necessário, ela incorre em um custo de oportunidade. O custo de oportunidade do caixa excedente (mantido em moeda ou em depósitos bancários) é a renda de juros que poderia ser ganha em aplicações financeiras, como um investimento em títulos negociáveis ou fundos de investimento em renda fixa.

Dado o custo de oportunidade de reter caixa, por que uma empresa reteria caixa excedente ao saldo médio mínimo negociado com um banco? A resposta é que um saldo de caixa deve ser

mantido para fornecer a liquidez necessária para as transações – o pagamento de contas. Se a empresa mantiver um saldo de caixa muito baixo, poderá ficar sem dinheiro. Se isso acontecer, talvez a empresa tenha de levantar caixa a curto prazo. Isso pode envolver, por exemplo, o resgate de aplicações financeiras ou de títulos negociáveis ou a tomada de empréstimos bancários.

A venda de títulos negociáveis ou o resgate de aplicações financeiras pode ter baixo custo, mas empréstimos envolvem diversos custos. Além disso, como já discutimos, manter saldos de caixa tem um custo de oportunidade. Para determinar o saldo de caixa apropriado, a empresa deve avaliar os benefícios de manter saldos de caixa em relação aos custos disso. Discutiremos esse assunto com maiores detalhes nas seções seguintes.

Gestão de caixa *versus* gestão da liquidez

Antes de continuarmos, devemos observar que é importante distinguir entre a gestão do caixa e um assunto mais geral, a gestão da liquidez. A distinção é fonte de confusão porque a palavra *caixa* é usada na prática de duas maneiras. Antes de mais nada, ela tem seu sentido literal, o caixa real, ou seja, o dinheiro disponível para uso. Entretanto, os gestores financeiros com frequência usam a palavra para se referir à soma da posição de caixa com a posição de investimentos financeiros e títulos negociáveis da empresa. Às vezes, esses títulos e investimentos são chamados de *equivalentes de caixa* ou *quase caixa*. Em nossa discussão sobre as situações de caixa da Ford e da GE no início do capítulo, por exemplo, o que descrevíamos na verdade era a soma do caixa total junto com o total dos equivalentes de caixa.

Indo direto ao ponto, a distinção entre gestão da liquidez e gestão do caixa é: a gestão da liquidez diz respeito à otimização da quantidade de ativos líquidos que uma empresa deve ter, sendo um aspecto particular das políticas da administração de ativos circulantes que discutimos no capítulo anterior. A gestão do caixa está muito mais relacionada à otimização dos mecanismos de cobrança e de desembolso de caixa, e esse é o assunto no qual nos concentramos basicamente neste capítulo.

Reservas bancárias

Um conceito fundamental para entendermos a administração do caixa no Brasil é o conceito de *reservas bancárias*. Reservas bancárias são os recursos imediatamente disponíveis mantidos pelos bancos junto ao Banco Central do Brasil. Todas as transações entre bancos e as transações bancárias de organizações não financeiras e pessoas físicas são realizadas mediante a transferência de titularidade de reservas. Todos os pagamentos e recebimentos somente se realizam por meio de reservas bancárias. A característica principal das reservas bancárias é que elas constituem recursos imediatamente disponíveis.

No dia a dia da atividade bancária, os bancos fazem e recebem pagamentos, concedem empréstimos e tomam dinheiro no mercado, e as entradas e saídas de caixa decorrentes dessas atividades podem fazer que alguns bancos apresentem sobra de reservas enquanto outros tenham falta de reservas.

Para buscar ajustar sua posição de reservas no Banco Central, os bancos negociam reservas entre si de forma a suprir necessidades opostas. De um modo geral, o sistema estará sempre com sobras ou falta de reservas. Nessa situação, o Banco Central atua para que as taxas de juros não oscilem para cima ou para baixo por causa das faltas e sobras de caixa no sistema e, dessa forma, ele tem o poder de influenciar a taxa de juros do mercado de reservas bancárias.[†]

As taxas Selic e DI

A taxa de juros de mercado que o Banco Central tem condições de influenciar diretamente é a *taxa Selic*. O nome Selic decorre de todas as transações com títulos públicos serem realizadas em reservas imediatamente disponíveis e de o registro dessas transações ser realizado pelo

[†] N. de R.T.: Ver processo de definição da taxa de juros, disponível em: http://www.bcb.gov.br/htms/relinf/port/1999/06/ri199906anp.pdf.

RESPOSTA DA REVISÃO DO CAPÍTULO E DO TESTE DE CONHECIMENTOS

19.1 O *float* de desembolso é de 7 dias × \$3.000 = \$21.000. O *float* de cobrança é de 2 dias × (−\$1.700) = −\$3.400. O *float* líquido é de \$21.000 + (−\$3.400) = \$17.600. Em outras palavras, em determinado momento, a empresa tem cheques não resgatados pendentes no valor de \$21.000. Ao mesmo tempo, ela tem cobranças não realizadas de \$3.400. Assim, o saldo contábil da empresa é \$17.600 menor do que seu saldo disponível para um *float* líquido positivo de \$17.600.

REVISÃO DE CONCEITOS E QUESTÕES INSTIGANTES

1. **Administração de caixa [OA3]** É possível que uma empresa tenha muito caixa? Por que os acionistas se importariam se uma empresa acumulasse grandes quantidades de caixa?
2. **Administração de caixa [OA3]** Quais opções estão disponíveis para uma empresa se ela acreditar que tem muito caixa? E acreditar que tem pouquíssimo caixa?
3. **Problemas de agência [OA3]** Acionistas e credores estão de acordo sobre a quantidade de caixa que deve ser mantida por uma empresa?
4. **Motivações para reter caixa [OA3]** Na abertura do capítulo, discutimos as situações de enormes saldos de caixa em várias empresas. Por que essas empresas reteriam esses valores tão grandes em caixa?
5. **Gestão de caixa *versus* gestão da liquidez [OA3]** Qual é a diferença entre a gestão de caixa e a gestão da liquidez?
6. **Investimentos de curto prazo [OA3]** Por que uma empresa não deve investir o caixa excedente por um prazo de poucos dias em uma carteira formada por títulos de longo prazo?
7. ***Floats* de cobrança e de desembolso [OA1]** O que uma empresa preferiria: um *float* de cobrança líquido ou um *float* de desembolso líquido? Por quê?
8. ***Float* [OA1]** Suponha que uma empresa tenha um saldo contábil de \$2 milhões. Ao consultar o caixa eletrônico, o administrador de caixa descobre que o saldo no banco é de \$2,5 milhões. Qual é a situação aqui?
9. **Investimentos de curto prazo [OA3]** Para cada um dos títulos negociáveis de curto prazo apresentados aqui, forneça um exemplo das possíveis desvantagens que o investimento tem para atender aos objetivos de administração de caixa de uma empresa:
   a. Títulos do Tesouro brasileiro.
   b. Ações.
   c. Certificado de depósito bancário.
   d. Notas promissórias (título mobiliário).
   e. Certificado de crédito imobiliário.
   f. Contratos de armazenagem.
10. **Problemas de agência [OA3]** Às vezes, afirma-se que o caixa excedente retido por uma empresa pode agravar os problemas de agência (discutidos no Capítulo 1) e, de modo geral, reduzir os incentivos para a maximização do valor para o acionista. Como você analisaria essa questão?
11. **Uso do caixa excedente [OA3]** Uma opção que uma empresa geralmente tem com caixa excedente é pagar seus fornecedores mais rapidamente. Quais são as vantagens e as desvantagens desse tipo de uso do caixa excedente?
12. **Uso do caixa excedente [OA3]** Outra opção normalmente disponível é reduzir a dívida pendente da empresa. Quais são as vantagens e as desvantagens desse uso do caixa excedente?
13. ***Float* [OA1]** Uma prática comum que infelizmente ocorre ainda nos Estados Unidos funciona mais ou menos assim (aviso: não tente fazer isso no Brasil): suponha que você

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Tirole

Jean Tirole - The Theory of Corporate Finance-Princeton Univ

- Fixed-investment model (baseline): Ch. 3, §3.2.1 (p. 115–116)
- Maturity of liabilities: liquidity shocks, cutoff, credit lines, deepening: Ch. 5, §5.2 (p. 201–207)
- Liquidity–scale tradeoff with two shocks: Ch. 5, §5.3.1 (p. 207–209)
- Incomplete hedging with serially correlated profits: Ch. 5, §5.4.2 (p. 216–220)
- Free cash flow and optimal claim maturity: Ch. 5, §5.6 (p. 225–228)

of the assets in the case of failure. We show that it is optimal for investors to have priority in the case of default and for the entrepreneur to be the residual claimant (Chapter 5 will investigate another feature of debt, namely, the borrower's promise to pay fixed amounts to investors as a going concern, i.e., before liquidation; and Chapter 10 will connect debtholders' control rights with their cash-flow rights).

By focusing on a simple model of credit rationing, we do not do full justice to the corporate finance literature, which has developed a wide array of models with a similar flavor. For the sake of completeness, the supplementary section studies three broad classes of models that, through more sophisticated modeling, have aimed at deriving an interpretation of the "leftover claim" of outside investors as a standard debt claim.

3.2 The Role of Net Worth: A Simple Model of Credit Rationing

3.2.1 The Fixed-Investment Model

Variants of the following entrepreneurial model[2] will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. (We will later explore the possibility that these assets be illiquid. For example, they might be equipment or premises that are needed for the implementation of the project.) The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. (We will later observe that we can ignore the possibility that the entrepreneur consumes some of the cash and borrows more than $I - A$.)

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit");[3] or, equivalently, the entrepreneur chooses between a project with a high probability of success and another project which ceteris paribus she prefers (is easier to implement, is more fun, has greater spinoffs in the future for the entrepreneur, benefits a friend, delivers perks, is more "glamorous," etc.) but has a lower probability of success.[4]

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur.[5] In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral.[6] For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0.[7] The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. That is, we have in mind that several prospective lenders compete for issuing a loan to the borrower, and that, if the most attractive loan offer made a positive profit, the borrower could turn to an alternative lender and offer

2. This specific model is taken from Holmström and Tirole (1997). But its main idea can be found in various forms in many anterior papers.

3. See Exercise 3.20 for the continuous-effort version of the model.

4. Note that, for simplicity, we treat the entrepreneur as a unitary actor. There is an interesting question as to how moral hazard and incentives propagate down within the corporate hierarchy. Pagano and Volpin (2005) assume that benefits accrue to all company insiders, and not only to managers; in their model, managers need workers' cooperation to produce and therefore share benefits with employees.

5. For example, in the biotechnology alliance financing discussed in Section 2.4.2, the private benefit might be the entrepreneur's benefit from working on other projects (with other partners or on her own). The shift in attention then reduces the probability of success of the project under consideration.

6. Exercise 3.2 generalizes this analysis to allow for entrepreneurial risk aversion. Exercise 3.12 considers risk-averse investors.

7. The investors have rate of time preference equal to 0, which is also the market rate of interest.

to switch for a slightly lower interest rate.[8] We use the plural "lenders" even though a single lender may turn out to finance the entire loan, because we want to emphasize that lending is a passive and anonymous activity in the theories reviewed in Part II.

Let us turn to the loan contract. A contract first stipulates whether the project is financed.[9] If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure (the gross payoffs are the *ex post* monetary payoffs and take no account of past investments and private benefit). Intuitively, there is no point in specifying a positive transfer from the lenders to the borrower, as such a transfer can only weaken incentives, while it has no insurance benefit under risk neutrality. This property will be proved more rigorously and is here taken for granted. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders.[10] To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

The zero-profit constraint for the lenders can be written as

$$p_H R_l = I - A,$$

assuming that the loan agreement induces the borrower to behave (which under our assumptions will be the case). The rate of interest ι is given by

$$R_l = (1+\iota)(I-A) \quad \text{or} \quad 1+\iota = 1/p_H.$$

So, unless $p_H = 1$, the nominal rate of interest ι reflects a default premium and exceeds the expected rate of return (called r in Part VI and here normalized to 0) demanded by investors.

We summarize the timing in Figure 3.1.

We assume that the project is viable only in the absence of moral hazard. That is, the project has

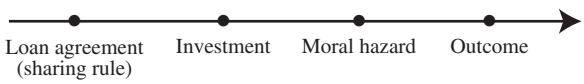


Figure 3.1

positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \tag{3.1}$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \tag{3.2}$$

It is easy to see that inequality (3.2) implies that no loan that gives an incentive to the borrower to misbehave will be granted. Indeed, rewrite (3.2) as

$$[p_L R_l - (I-A)] + [p_L R_b + B - A] < 0.$$

So, in the case of misbehavior, either the lenders must lose money in expectation, or the borrower would be better off using her cash for consumption, or both.

3.2.2 The Lenders' Credit Analysis

Because the project has negative NPV in the case of misbehavior, the loan agreement must be careful to preserve enough of a stake for the borrower in the enterprise. The borrower faces the following tradeoff once the financing has been secured: by misbehaving, she obtains private benefit B, but she reduces the probability of success from p_H to p_L. Because she has stake R_b in the firm's income (she receives R_b in the case of success and 0 in the case of failure), the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \tag{IC$_b$}$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H\left(R - \frac{B}{\Delta p}\right).$$

8. See Exercise 3.13 for the extension of the model to lender market power.

9. "Random financing" contracts, in which the borrower brings equity in exchange for a probability between 0 and 1 of being financed may in some cases be optimal when the investment size is fixed (as it is here) or more generally in the presence of indivisibilities or increasing returns to scale (see Exercise 3.1). For simplicity, we focus on deterministic contracts.

10. The lenders' net payoff is thus $R_l - (I-A)$ in the case of success, and $-(I-A)$ in the case of failure. The borrower's net payoff is thus $R_b - A$ in the case of success, and $-A$ in the case of failure, to which, in both cases, must be added a private benefit B if shirking occurs.

reflects managerial decisions and not solely extraneous uncertainty. For incentives reasons, the amount of liquidity available to the firm should then increase with the realized cash flow; that is, reinvestment should be sensitive to cash flow (which corresponds well to the empirical tests of the sensitivity of investment to cash flow, which are performed on ongoing entities and demonstrate a positive association between reinvestment and cash flow). There is, however, no theoretical ground for assuming that this sensitivity decreases with the strength of the firm's balance sheet.

While Sections 5.2–5.4 emphasize the point that the capital market may *ex post* rationally, but inefficiently, deny funds to the firm, Section 5.5 also studies the opposite phenomenon of a capital market that is too lenient with the borrower. When the liquidity shock is endogenous, that is, depends on the borrower's behavior, it may be optimal to let the firm fail even for moderate liquidity shocks. The prospect of failure then acts as a disciplining device for the borrower, and induces her to better control liquidity needs. Once the need for liquidity accrues, however, it may no longer be optimal for the capital market to adhere to this tough stance. Indeed, if the expected return from continuation exceeds the agency cost, the borrower can successfully renegotiate the initial agreement and obtain more funds. This is the phenomenon of the soft budget constraint. We then show how the soft-budget-constraint problem may arise whenever more general news about poor past performance accrues at the intermediate stage.

Following Easterbrook (1984) and Jensen (1986, 1989), Section 5.6 focuses on cash-rich firms, defined as firms with cash inflows exceeding their efficient reinvestment needs or opportunities. Such firms have excess liquidity that must be "pumped out" in order not to be used on wasteful projects, unwarranted diversifications, perks, and so forth. Jensen's (1989) list of industries with potential free-cash-flow problems includes steel, chemical, television and radio broadcasting, brewing, tobacco, and wood and paper products.

Overall, the liquidity-shortage and free-cash-flow problems are two sides of the same coin. The key issue in the design of long-term financing is to ensure that, at intermediate stages, the right amount of money is available for the payment of operating expenses and for reinvestment and the right amount is paid out to investors. Whether this results in a net inflow (the liquidity-shortage case) or outflow (the free-cash-flow case) is important for the comprehension of corporate financing, but is a pure convention as far as economic principles are concerned. And, indeed, we merely reinterpret the liquidity-shortage model in order to obtain its flip side, the free-cash-flow model.

The exposition in this chapter is based in part on joint work (in particular, Holmström and Tirole 1998, 2000) and numerous discussions with Bengt Holmström.

5.2 The Maturity of Liabilities

5.2.1 Basics

We depart from the previous sole focus on solvency by introducing the possibility that, during the implementation of the project (of size I), the firm be hit by an adverse shock and be required to plow in some extra cash in order to be able to pursue the project. A firm has two ways of facing urgent liquidity needs if it lacks funds (either because it generates no cash in the short run (a "cash-poor firm") or because it generates enough income in the short-run to cover reinvestment needs ("cash-rich firm") but pays out part or all of this income and therefore has limited retentions). The first is to secure some source of cash before the liquidity shock occurs. For example, the firm may "overborrow" and keep liquid assets such as Treasury bills on its balance sheet in order to be able to absorb the shock by selling these assets when needed. Alternatively, the firm may secure a line of credit with a lender (usually a bank). In contrast, the second approach consists in waiting for the shock to occur to start raising funds.

As explained in the introduction, the wait-and-see approach generates excessive liquidity problems. That is, there are situations where the firm would be rescued under an optimal contract but neither initial lenders nor new lenders want to participate even in a coordinated rescue. This is due to the fact that the borrower's stake is incompressible, that is, a concession by the borrower (in the form of a reduction of her stake) creates moral hazard and is

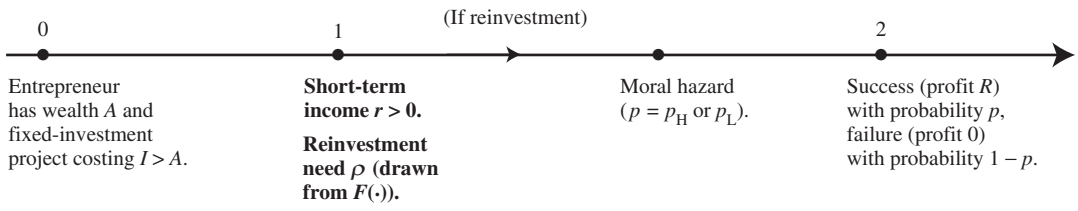


Figure 5.1

unacceptable to lenders. So, the lenders do not internalize the loss incurred by the borrower when the project is stopped, and yet the borrower is unable to propose a concession to induce them to internalize this externality.

Consider the setup of Section 3.2, except that there is an intermediate date at which income accrues and some reinvestment need is realized. As indicated in Figure 5.1, the entrepreneur at date 0 has wealth A and borrows $I - A$, where I is the fixed cost of investment.

At date 1, the investment yields deterministic and verifiable income $r \geqslant 0$. Continuation, though, requires reinvesting an amount ρ, where ρ is *ex ante* unknown and has cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0, \infty)$. The realization of ρ is learned at date 1. Note that we here assume that the date-1 income is deterministic while the reinvestment need is random. The important assumption is that at least one of the two is random.

If the firm does not reinvest ρ, then the firm is liquidated. The liquidation value is 0. If the firm reinvests ρ, then the firm yields, at date 2, R with probability p and 0 with probability $1 - p$, where $p = p_H$ if the entrepreneur behaves (and then gets no private benefit) and $p = p_L = p_H - \Delta p$ if the entrepreneur misbehaves (in which case she receives private benefit B).

The entrepreneur and the investors are risk neutral, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Thus, the model is nothing but an extension of the basic fixed-investment one in Section 3.2. We have just added an intermediate income r and a reinvestment need ρ (the bold type in Figure 5.1). (Put differently, the model of Section 3.2 corresponds to the special case $r = 0$ and F being a spike at $\rho = 0$.) We assume that there exists in the economy a store of value that yields the consumers' rate of interest (0 here). That is, 1 unit invested at date 0 delivers a return of 1 unit at date 1 (Chapter 15 will investigate the reasonableness of this assumption). We now give a *heuristic* description of the optimal contract.

Suppose in a first step that the initial contract can specify whether the firm continues or liquidates for each value of ρ (as we will see, it actually does not matter whether the realized value of ρ is verifiable, as long as there is no use that can be made of the date-1 cash flow besides reinvesting it and distributing it to investors). Intuitively, it is optimal to continue whenever it is cheap to do so:

$$\rho \leqslant \rho^*,$$

where ρ^* is a cutoff.

As is now familiar to the reader, competition among investors deprives them of a surplus, and so the borrower's utility is equal to the NPV. Assuming, as usual, that the optimal contract induces the high effort in the case of continuation and noting that the probability of continuation is $\Pr(\rho \leqslant \rho^*) = F(\rho^*)$, the borrower's net utility is

$$U_b(\rho^*) = [r + F(\rho^*)p_H R] - \left[I + \int_0^{\rho^*} \rho f(\rho)\, d\rho\right],$$

where the first bracket represents expected revenue and the second bracket total investment (initial investment plus expected reinvestment).

Ensuring good behavior in the case of continuation suggests giving to the entrepreneur, at date 2, R_b in the case of success and 0 in the case of failure, where

$$(\Delta p)R_b \geqslant B.$$

Furthermore, there is no loss of generality in assuming that the entrepreneur receives nothing at date 1. Suppose she receives $r_b > 0$. Then the contract could eliminate this short-term compensation

and increase R_b by δR_b, so that the expected total reward remains constant: $F(\rho^*)p_H\delta R_b = r_b$. If anything, this substitution alleviates moral hazard in the case of continuation. And the suppression of the date-1 compensation does nothing to the date-1 income (which we took to be exogenous, an assumption we relax in Section 5.5).

The pledgeable income, $\mathcal{P}$, deflated by the investors' initial outlay, $I - A$, is therefore

$$\mathcal{P}(\rho^*) - (I - A) = \left[r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right]\right] - \left[I + \int_0^{\rho^*} \rho f(\rho)\,d\rho - A\right],$$

since the entrepreneur no longer has cash and so the reinvestment must be paid out of either the investors' pocket or date-1 revenue.

Taking derivatives in U_b and $\mathcal{P}$, the key insights are as follows:

- The NPV (U_b) is increasing in the cutoff ρ^* as long as $\rho^* < p_H R$, and is decreasing thereafter. Intuitively, one would want to salvage an investment when the cost, ρ, of a rescue is smaller than the expected payoff, $p_H R$, of continuing.
- By contrast, the pledgeable income increases with ρ^* for $\rho^* < p_H(R - B/\Delta p)$ and decreases thereafter. This is again intuitive: investors have to bear the cost, ρ, of salvaging the investment and can put their hands on at most $p_H(R - B/\Delta p)$ given that the entrepreneur must be given incentives to behave in the case of continuation.

We are then led to consider three cases. Depending on the strength of the balance sheet, there may be (i) an efficient amount of liquidation, (ii) an over-optimal amount of liquidation to satisfy investors, or (iii) no funding at all:

(i) $\mathcal{P}(p_H R) \geqslant I - A$.

In this case, the "first-best cutoff" $\rho^* = p_H R$, which maximizes U_b, leaves sufficient income to investors. The contract then specifies, say, no compensation r_b at date 1 for the entrepreneur, and a reward R_b in the case of continuation and success at date 2.[4]

(ii) $\mathcal{P}(p_H R) < I - A \leqslant \mathcal{P}\left(p_H\left(R - \frac{B}{\Delta p}\right)\right)$.

The optimal contract then specifies[5] $r_b = 0$ and $R_b = B/\Delta p$. The entrepreneur receives nothing at the intermediate date and, in the case of continuation, receives the lowest compensation, $R_b = B/\Delta p$, that is incentive compatible. Intuitively, the entrepreneur can be paid in two currencies: cash and continuation. Cash payments are just transfers and do not affect the NPV (as long as incentive compatibility obtains); as long as $\rho < p_H R$, continuation is a more efficient currency since continuation increases the NPV.

The cutoff $\rho^* \in [p_H(R - B/\Delta p), p_H R]$ is then given by[6]

$$r + F(\rho^*)\left[p_H\left(R - \frac{B}{\Delta p}\right)\right] = I + \int_0^{\rho^*} \rho f(\rho)\,d\rho - A.$$

Figure 5.2 illustrates the determination of the cutoff in this region. The pervasive logic of credit rationing applies not only to the choice of initial investment, but also to the continuation decision. In order to be able to invest more *ex ante*, the borrower accepts a level of reinvestment below the *ex post* efficient level ($\rho^* < p_H R$). The intuition is that, because incentives must be preserved, the borrower cannot pledge to the lenders the entire benefit of the reinvestment decision. Also, ρ^* exceeds the per-unit pledgeable income $p_H(R - B/\Delta p)$, which is the level that maximizes the borrowing capacity. A small increase in ρ^* at that level induces only a second-order decrease in

4. More generally, r_b and R_b are given by the investors' breakeven condition:

$$r - r_b + F(p_H R)[p_H(R - R_b)] = I + \int_0^{p_H R} \rho f(\rho)\,d\rho - A,$$

as long as $R_b \geqslant B/\Delta p$ and $r_b \geqslant 0$.

5. Here there is no indeterminacy. A positive r_b reduces ρ^*, which in turn reduces U_b.

6. An early paper emphasizing the role of the insiders' stake and the absence of maximization of the firm's value to investors in the optimal choice of an interim policy, such as continuation and restructuring, is Chang (1992). In that paper, the interim decision consists in restructuring the firm, thereby imposing a cost on insiders. It is shown that restructuring occurs less often than it would if investors had noncontingent control rights over the restructuring decision and therefore chose to restructure the firm whenever this increased the firm's interim value.

Here, abandoning the project (the analog of restructuring in Chang's paper) maximizes the investors' interim value whenever $\rho > p_H[R - B/\Delta p]$. However, abandoning imposes a cost on the entrepreneur, namely, the loss of rent $p_H B/\Delta p$. The firm continues in a broader set of circumstances than would maximize the investors' interim value, in the same way as restructuring occurs less often than would be the case if one maximized the investors' interim value in Chang's paper. Chang studies the implications for the allocation of control rights. We focus on those for liquidity management.

See also Dasgupta and Sengupta (2005) for a recent contribution to this literature.

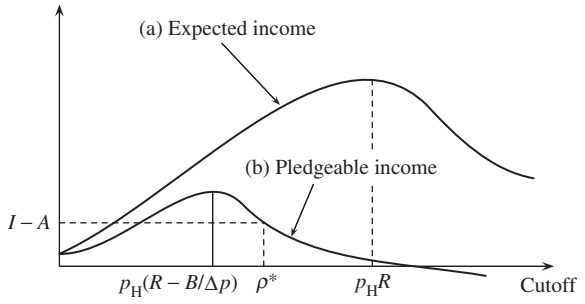


Figure 5.2 Optimal continuation policy:
(a) $U_b + I = r + F(\rho^*)[p_H R] - \int_0^{\rho^*} \rho f(\rho)\, d\rho$;
(b) $\mathcal{P}(\rho^*) = r + F(\rho^*)[p_H(R - B/\Delta p)] - \int_0^{\rho^*} \rho f(\rho)\, d\rho$.

debt capacity, and a first-order gain in the efficiency of *ex post* refinancing.

(iii) $\mathcal{P}\left(p_H\left(R - \dfrac{B}{\Delta p}\right)\right) < I - A.$

In this case, funding is not feasible. The value ρ^* that maximizes the pledgeable income ($\rho^* = p_H(R - B/\Delta p)$) does not suffice to compensate the investors for their initial outlay.

5.2.2 Term Structure of Cash-Rich Firms

Let us define a cash-rich firm as one that is meant to disgorge money at the intermediate stage: $r > \rho^*$ (in particular, $r \geqslant p_H R$ suffices to ensure that the firm is cash rich). The optimal contract can be implemented through a combination of *short-term debt*,

$$d = r - \rho^*,$$

and *long-term debt* (to be paid in the case of continuation),[7]

$$D = R - \frac{B}{\Delta p}.$$

We thus obtain a simple theory of *maturity structure*. Note further that as the strength of the balance sheet, as measured by the value of A, changes, only ρ^* changes. In particular, as A increases, ρ^* also increases (see region (ii) in Section 5.2.1; it increases only weakly in region (i)), and so d decreases. Conversely, *a weak balance sheet implies a short maturity structure* (d large).

This helps us to understand why *highly indebted firms are more likely to borrow on a short-term basis.* Highly leveraged firms can be viewed as firms with a weak balance sheet,[8] and so must accept shorter maturities.

Similarly, if we added another margin of concession in the form of costly collateral pledging (thus combining this section with the modeling in Section 4.3), one would find that firms with weak balance sheets borrow on a short-term and secured basis.

Discussion. While we emphasize the short-term debt interpretation, this payment can actually be interpreted either as a short-term debt as in Jensen (1986) or as a dividend as in Easterbrook (1984). Note, though, that the dividend interpretation must be accompanied by a covenant concerning maximal dividend distribution.[9] Otherwise, investors would want to pay dividends up to $r - \rho_0 > d$, where $\rho_0 = p_H(R - B/\Delta p)$, in order to prevent the entrepreneur from reinvesting whenever the liquidity shock exceeds the date-1 pledgeable income ρ_0. With this interpretation, we see that covenants specifying maximal amounts of dividends serve to protect the *entrepreneur* against excessive liquidation.[10]

7. Here, long-term debt and equity are equivalent. To obtain three different claims (short-term debt, long-term debt, equity), one can proceed as in Chapter 3 and introduce a leftover value in the case of failure at date 2.

8. Suppose that the firm already owes D_0 at date 2. The income in the case of success is then $R - D_0$. The analysis above shows that in the constrained region (ii), ρ^* decreases as D_0 increases. And so the short-term debt d increases.

Things get more complex when the initial debt is short-term debt (d_0). Then the disposable short-term debt revenue becomes $r - d_0$. The cutoff ρ^* decreases with d_0. *Total* short-term debt ($d_0 + d$) increases with d_0, but the sign of the impact on *new* short-term debt d depends on distributional assumptions.

This analysis presumes, as in Section 3.3, that initial short-term (d_0) or long-term (D_0) debts are not renegotiated. The analysis is different if initial debtholders can be brought to the bargaining table, but the general point that their presence weakens the firm's balance sheet remains.

9. In practice, dividends may also be limited because managers have some control over their level (this alternative story is more complex to analyze than the covenant one because it relies on the drivers' of managerial "real authority" (see Chapter 10 for the concept of real authority)).

10. This insight complements the standard, and important explanation for the existence of such covenants. As discussed in Chapter 2, they are usually viewed as protecting *creditors* against expropriation by the equityholders, who could use dividend distributions and share repurchases to leave long-term creditors with an "empty shell." In this part, we focus on the conflict between the entrepreneur and the securityholders, and so the introduction of conflicts among securityholders would serve no purpose.

This study focuses on the conflict between the entrepreneur and the investors concerning payments to investors out of cash-flow, without going into the details of whether the payment d must be interpreted as short-term debt or as a (constrained) dividend. That is, it is general enough to encompass the theories of Easterbrook and Jensen, but ought to be refined in order to motivate a diversity of securities. Note also that by predicting a fixed payment d, it does not do justice to the rich range of conditional payments observed in practice, that endow investors with more or less flexibility in pumping cash out of the firm: dividend, preferred dividend, putable securities, renegotiated short-term debt, short-term debt (we will return to this point in Section 5.6.2).

5.2.3 Credit Lines for Cash-Poor Firms

Suppose in contrast that the investment "takes a long time" to produce income. At the extreme, there is no short-term profit: $r = 0$.

Can the entrepreneur just "wait and see," that is, borrow I at date 0 in exchange for shares in the firm and return to the capital market at date 1 if need occurs? Let us thus assume that the entrepreneur does not plan her liquidity in advance and that the liquidity shock occurs at date 1. To raise cash on the capital market to pay ρ, the entrepreneur must issue new shares and thereby dilute historical investors.

Letting $\rho_0 \equiv p_{\rm H}(R - B/\Delta p)$, and to illustrate this dilution, suppose that the entrepreneur faces a liquidity shock $\rho = \frac{1}{2}\rho_0$. The value of external shares held by initial investors is equal to ρ_0. Suppose that the number of shares is doubled.[11] That is, as many shares are sold to new investors as already exist. So the value of each share is halved. The firm thereby raises $\frac{1}{2}\rho_0 = \rho$ in cash and can withstand the liquidity shock. Are initial investors willing to let themselves be diluted? The value of their shares is, of course, reduced to $\frac{1}{2}\rho_0$. But contemplate the alternative of liquidating the investment, under which the initial investors receive nothing! Thus, initial investors are willing to accept the dilution.[12]

Similarly, to meet a liquidity shock equal to $\frac{3}{4}\rho_0$, the firm must quadruple the number of shares, and so on. But there is an upper bound to this process: investors will never pay more than the firm is worth to them. Hence, even in a frictionless capital market, the firm cannot raise more than ρ_0. Going back to the capital market at date 1 then at best allows the firm to withstand a liquidity shock of magnitude

$$\rho \leqslant \rho_0 = p_{\rm H}\left(R - \frac{B}{\Delta p}\right).$$

Because the optimal financing arrangement specifies[13]

$$p_{\rm H}\left(R - \frac{B}{\Delta p}\right) < \rho^* \leqslant p_{\rm H}R,$$

the entrepreneur must secure a line of credit or hoard liquidity in order to face the date-1 liquidity shock.[14] We will shortly describe how to do so, but there are basically two alternatives and combinations thereof: a credit line or liquid assets of magnitude ρ^* with no right to dilute existing claimholders by issuing new claims at date 1 (so the entrepreneur borrows $I + \rho^*$); or a smaller credit line or amount of liquid assets, equal to $[\rho^* - p_{\rm H}(R - B/\Delta p)]$ with a right to dilute claimholders as needed to ensure continuation. Either way, the entrepreneur must plan liquidity management.

The optimum can be implemented by a nonrevokable line of credit granted by, say, one of the lenders (a bank) at level ρ^*. It is important that this line of credit be nonrevokable (in a broad sense: see below). Otherwise the lender would have an incentive not to abide by his promise to rescue the firm if $\rho > \rho_0$, that is, if the liquidity shock exceeds the date-1 pledgeable income ρ_0. In practice, lenders often prefer to keep discretion over the extension of credit by making the line revokable, or delivering promises such as "comfort or highly confident letters," which are legally hard to enforce and are only a moral promise to provide credit. This discretion potentially has a cost to the borrower, as, whenever $\rho_0 < \rho < \rho^*$, the lender would like to renege on his promise to provide funds to the firm unless he tries to maintain a reputation for "fairness" by extending credit even when this is not strictly profitable for him (see Boot

11. Including for internal shares, so as to keep the entrepreneur's stake $R_{\rm b}$ in success constant and therefore preserve incentive compatibility.

12. Note the analogy with the incentives for debt forgiveness when there is a debt overhang (see Chapter 3).

13. Except in the nongeneric case where $\mathcal{P}(p_{\rm H}(R - B/\Delta p)) = I - A$.

14. Note that if $A > 0$, the hoarding of liquidity can in part come from the retention of A (this is a matter of accounting).

et al. 1987, 1993). In practice, a bank may offer a formally revokable line of credit, but have a reputation for abiding by its promise unless the borrower has behaved in an egregious way that yet was not ruled out by a negative covenant.

We also implicitly assume that there is no concern over the lender's *ability* (as opposed to *willingness*) to abide by his commitment. However, the lender may himself face liquidity and solvency problems in the future. In practice, only well-capitalized and safe institutions are able to make a firm promise of this type (banks and some other financial institutions obviously have a comparative advantage in doing so, due to the close monitoring of their solvency and liquidity by the regulators as well as, at least for large ones, an explicit or implicit backing of their on- and off-balance-sheet liabilities by the state).

Remark (capital market frictions). Note here that the suboptimality of reinvestment under the wait-and-see policy is independent of the debt-overhang phenomenon discussed in Section 3.3. Indeed, the assumption that liquidity shocks below ρ_0 can be withstood through the dilution of existing claims implies either that lending is concentrated among a few lenders, or that the initial agreement is structured so as to facilitate renegotiation,[15] or else that the entrepreneur receives rights to dilute existing claims by issuing senior claims (as in Hart and Moore (1995)). If some claims proved difficult to renegotiate, the firm would be able to raise even less than ρ_0 by turning to the capital market at date 1, and its demand for liquidity would be even higher than that derived here.

Remark (renegotiation). Could this line of credit be renegotiated to the parties' mutual advantage once the fraction ρ is realized? First, note that if $\rho \leqslant \rho^*$, then *a fortiori* $\rho < p_H R$ and therefore it is *ex post* efficient to continue; so there is no scope for a renegotiation in which the lender would compensate the borrower for not using the credit line, as this renegotiation would reduce total surplus and therefore at least one of the parties would be strictly better off not renegotiating. Second, could the two parties both benefit from an increase in the line of credit to ρ when $\rho^* < \rho \leqslant p_H R$? Even though this increase would yield the *ex post* efficient reinvestment policy, there is no way for the borrower to compensate the lender, again because the borrower's stake is incompressible. One can show that the lender will turn down any request for an increase in the credit line.[16] So will any alternative lender (other lenders may have even less incentives to refinance, because unlike the initial lender they do not have a vested stake to lose).

Remark (role played by uncertainty about liquidity needs). We can now explain why *ex ante* uncertainty about the liquidity need is a key ingredient of the demand for liquidity. Suppose, in contrast, that ρ is deterministic. If $\rho \geqslant \rho_0 = p_H(R - B/\Delta p)$, then investors do not want to lend at date 0, since they know that they will have to cover at date 1 a liquidity shock that exceeds the income that can be pledged to them in period 2. If $\rho < \rho_0$, then the firm is always solvent at date 1, in that new claims can be issued at date 1 (that partially dilute existing ones) in order to meet the liquidity shock and continue; hence, there is no need to hoard reserves.

Again, a good way of thinking about this issue is in terms of insurance. A high liquidity shock is similar to an illness or an accident, and a low liquidity shock is similar to the absence of such a mishap. There is no scope for insurance if it is known in advance whether there will be an illness or an accident.

5.2.4 A Reinterpretation: Growth Prospects

In the basic model, the firm is liquidated if it does not meet the liquidity shock. In a straightforward reinterpretation, it continues as is, but cannot take advantage of a profitable growth opportunity if it does not come up with enough cash to reinvest.

Suppose that, at date 1, the firm still receives deterministic income r, but, in the absence of cash reinjection at date 1, continues and succeeds with probability $p = p_H$ or p_L, depending on whether the entrepreneur behaves or misbehaves at date 1. At date 1, though, the firm can raise its date-2 expected profit by reinvesting. One way of formalizing

15. See Section 5.5.3 for a discussion of factors hindering and facilitating renegotiation of claims.

16. More formally, the lender turns down the request because $\rho > \rho^* > \rho_0$.

this is to assume that the payoffs in the cases of success and failure remain R and 0, respectively, but the probability of success in the case of reinvestment becomes $p+\tau$, where $\tau>0$ and $p=p_{\mathrm{H}}$ or $p=p_{\mathrm{L}}$, depending on whether the entrepreneur behaves or misbehaves. This separable form is handy as it implies that the entrepreneur's incentive compatibility constraint is not affected by the reinvestment:

$$(p_{\mathrm{H}}+\tau)R_{\mathrm{b}} \geqslant (p_{\mathrm{L}}+\tau)R_{\mathrm{b}}+B \quad \Longleftrightarrow \quad (\Delta p)R_{\mathrm{b}} \geqslant B.$$

The reinvestment cost ρ is drawn at date 1 from the cumulative distribution function $F(\rho)$ with density $f(\rho)$ on $[0,\infty)$.

It is clearly optimal to reinvest if and only if ρ is below some cutoff ρ^*. The entrepreneur's utility, equal to the NPV, is

$$U_{\mathrm{b}}(\rho^*) = [r+[p_{\mathrm{H}}+F(\rho^*)\tau]R]-\left[I+\int_0^{\rho^*}\rho f(\rho)\,\mathrm{d}\rho\right].$$

As earlier, the interesting case arises when the firm is financially constrained but nonetheless can raise funds (a situation equivalent to that labeled region (ii) in Section 5.2.1); the cutoff is then given by the investors' breakeven condition:

$$r+[p_{\mathrm{H}}+F(\rho^*)\tau]\left(R-\frac{B}{\Delta p}\right) = [I-A]+\int_0^{\rho^*}\rho f(\rho)\,\mathrm{d}\rho$$

and

$$\tau\left(R-\frac{B}{\Delta p}\right) \leqslant \rho^* < \tau R.$$

The latter set of inequalities expresses the fact that reinvestment is first-best suboptimal ($\rho^*<\tau R$), but occurs whenever it boosts pledgeable income ($\rho^* \geqslant \tau[R-(B/\Delta p)]$).

In this model, growth opportunities are measured by the parameter τ. Let us look at the impact of growth opportunities on the maturity structure by differentiating the investors' breakeven condition:

$$\begin{aligned}\frac{\mathrm{d}(d)}{\mathrm{d}\tau} &= \frac{\mathrm{d}(r-\rho^*)}{\mathrm{d}\tau}\\ &= -\frac{F(\rho^*)}{f(\rho^*)}\frac{R-B/\Delta p}{\rho^*-\tau(R-B/\Delta p)} < 0.\end{aligned}$$

Thus, *firms with better growth opportunities should go for longer maturities*. Relatedly, there is substantial evidence that firms with growth opportunities have lower leverage ratios.[17]

17. See Section 2.5. Recall that equity here can be viewed as debt with a long maturity.

5.3 The Liquidity–Scale Tradeoff

The fixed-investment model is handy to illustrate the optimal term structure of debt for cash-rich firms and credit line for cash-poor ones. But, for other purposes, it is too simple, in that there is no other "margin" that the entrepreneur can trade off against liquidity. When, for example, investment size is variable, as we now assume, the entrepreneur faces a choice between a larger investment and more liquidity.[18]

This section focuses on cash-poor firms and extends the model of Section 5.2 to include a variable investment size in order to identify the liquidity–scale tradeoff (which also applies to cash-rich firms): the firm must sacrifice scale in order to benefit from more liquidity.

5.3.1 The Two-Shock Case

We consider the variable-investment model and add a liquidity shock at an intermediate stage. This liquidity shock amounts to a cost overrun that is proportional to the initial investment. To develop our intuition, let us begin with the case in which there are only two possible values for the (per-unit) liquidity shock: 0 with probability $1-\lambda$ and ρ with probability λ (see Figure 5.3). We will say that the firm is "intact" when it does not need to reinvest and "in distress" when it needs to reinvest ρ per unit of investment.

Except for this random shock, the model is identical to the variable-investment version of Section 3.4. Continuation (which is contingent on reinvesting ρI if the firm is in distress) is subject to moral hazard. The probability of success is p_{H} if the entrepreneur behaves and p_{L} if she misbehaves. The private benefit of misbehaving is BI. The project yields RI in the case of success and 0 in the case of failure. Note that we focus on policies that rescue either the entire investment or none of it in the case of distress.[19]

18. More generally, the entrepreneur would face a tradeoff between more liquidity and fewer control rights granted to investors (see Chapter 10), and so forth.

19. Quite generally, we could allow partial reinvestments. That is, a reinvestment $\rho x I$ allows the firm to salvage a fraction $x\in[0,1]$ of the investment. In this case, the private benefit of misbehaving, BxI, is proportional to the salvaged investment xI; and so is the profit RxI in the case of success. But it turns out that one can focus without loss of generality on policies that either rescue the entire investment ($x=1$) or rescue none ($x=0$) in the case of distress.

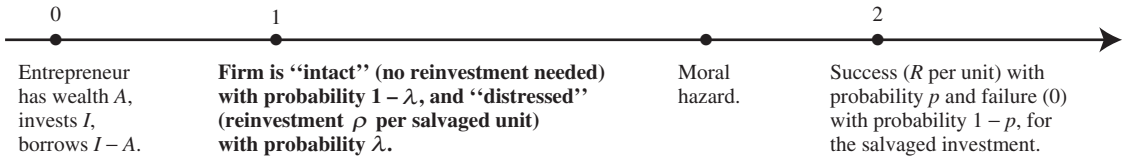


Figure 5.3

Let us assume that

$$\begin{aligned}\rho_0 &\equiv p_H\left(R - \frac{B}{\Delta p}\right)\\ &< c \equiv \min\left\{1 + \lambda\rho, \frac{1}{1-\lambda}\right\}\\ &< \rho_1 \equiv p_H R.\end{aligned}$$

This pair of inequalities (which boils down to $\rho_0 < 1 < \rho_1$ in the no-liquidity-shock case ($\lambda = 0$) of Section 3.4) will, as we will see, imply that investing has a positive NPV, but also that the entrepreneur is constrained in her borrowing.

In the case of continuation, the entrepreneur optimally receives 0 in the case of failure and R_b in the case of success, where R_b is large enough so as to incentivize her:

$$(\Delta p)R_b \geqslant BI.$$

As in Section 3.4, making this inequality an equality maximizes the pledgeable income and thereby the entrepreneur's borrowing capacity. This implies that under continuation, an expected amount $\rho_0 I$ goes to investors at date 2.

Let us compare the two policies.

(i) *Abandon the project in the case of distress.* If the project is abandoned in the case of distress, investors receive their expected income $\rho_0 I$ only when there is no shock, that is, with probability $1 - \lambda$. On the other hand, there is no reinvestment at date 1. Thus, when the entrepreneur has initial wealth A, the investors' breakeven constraint is

$$(1-\lambda)\rho_0 I = I - A,$$

yielding investment capacity,

$$I = \frac{A}{1-(1-\lambda)\rho_0}$$

(a generalization of formula (3.12) to the case $\lambda \geqslant 0$).

The entrepreneur's utility, equal to the NPV, is

$$U_b^0 = [(1-\lambda)\rho_1 - 1]I = \frac{(1-\lambda)\rho_1 - 1}{1-(1-\lambda)\rho_0}A$$

or

$$U_b^0 = \left[\left(\rho_1 - \frac{1}{1-\lambda}\right)\Big/\left(\frac{1}{1-\lambda} - \rho_0\right)\right]A.$$

Comparing this formula with that in the absence of a liquidity shock ($\lambda = 0$), the average cost of bringing 1 unit of effective or intact investment to date 2 is now $1/(1-\lambda)$ instead of 1, because the initial investment bears fruits only if there is no liquidity shock.

(ii) *Pursue the project even in the case of distress.* The decision to withstand the liquidity shock at date 1 has a cost and benefit. The cost is that the average cost of bringing 1 unit of investment intact to date 2 is $(1 + \lambda\rho)$ (the date-0 cost, 1, plus the expected date-1 reinvestment cost, $\lambda\rho$). The benefit is that the project is never abandoned. The borrowing capacity is given by

$$(1+\lambda\rho)I - A = \rho_0 I$$

or

$$I = \frac{A}{(1+\lambda\rho) - \rho_0}.$$

Similarly, the entrepreneur's utility (the NPV) is

$$U_b^1 = [\rho_1 - (1+\lambda\rho)]I$$

or

$$U_b^1 = \frac{\rho_1 - (1+\lambda\rho)}{(1+\lambda\rho) - \rho_0}A$$

(which, again, for $\lambda = 0$, boils down to formula (3.14′) in Section 3.4).

Thus, we find a similar formula as in the alternative policy, except that the average cost of effective investment is now $(1 + \lambda\rho)$.

The policy of withstanding the liquidity shock is optimal if and only if $U_b^1 \geqslant U_b^0$, or

$$1 + \lambda\rho \leqslant \frac{1}{1-\lambda},$$

which can be rewritten as

$$(1-\lambda)\rho \leqslant 1.$$

In words, it is optimal to withstand the liquidity

shock if

- it is low (ρ low),
- it is likely (λ high).

The first conclusion is obvious; but the second may be less so since a high probability of a liquidity shock increases both the benefit and the cost of withstanding it.

As in the case of a fixed investment size, we can draw the implications of this analysis for liquidity management. If the optimal policy is not to rescue the investment in the case of distress, nothing needs to be done at date 0 besides signing a contract and investing I. In contrast, if the optimal policy is to pursue the project even in the case of distress, the entrepreneur must be able to avail herself of the amount ρI if a shock occurs.

If $\rho > \rho_0$ (which is not inconsistent with the condition $(1-\lambda)\rho \leqslant 1$ obtained earlier), then liquidity necessarily must be planned in advance. Waiting exposes the firm to credit rationing at date 1. (As the analysis for a continuum of liquidity shock will demonstrate, this case is in a sense the "generic case.") For example, the firm may contract a credit line to the level of ρI with a bank; alternatively, it can contract for a credit line corresponding only to the shortfall $(\rho - \rho_0)I$ and also acquire the right to dilute initial investors (so as to obtain $\rho_0 I$). More on this in Section 5.3.3.

5.3.2 Continuum of Liquidity Shocks

We now generalize the analysis to a continuum of possible values for the liquidity shock. This continuous-investment, continuous-shock version will be used in the rest of the chapter.

After the (endogenous size) investment I is sunk at date 0 and before the borrower works on the project, some exogenous shock occurs at date 1 that determines a per-unit-of-investment level $\rho \in [0, \infty)$ of "cost overruns." That is, a cash infusion equal to ρI is needed in order for the project to continue. If ρI is not invested, the project is abandoned altogether and thus yields no income. As in Section 5.2, the fraction ρ is *a priori* distributed according to the continuous distribution $F(\rho)$ on $[0, \infty)$, with density $f(\rho)$. (As we already observed, the model of Section 3.4 is therefore a special case, with F being a spike at $\rho = 0$.)

Regardless of the required amount of the cash infusion, the project, if pursued, is still a project of size I, in that the income in the case of success is RI and the borrower's private benefit from misbehaving is BI. One cannot increase the size of the project after the initial stage.

The timing is summarized in Figure 5.4.

We assume that investment has positive NPV. That is, under a rule that specifies that the project is abandoned if and only if $\rho \geqslant \tilde{\rho}$ for at least some threshold $\tilde{\rho}$, the expected payoff per unit of investment is strictly positive. This positive-NPV condition under liquidity shocks is

$$\max_{\tilde{\rho}} \left\{ F(\tilde{\rho}) p_{\mathrm{H}} R - 1 - \int_0^{\tilde{\rho}} \rho f(\rho)\, \mathrm{d}\rho \right\} > 0. \tag{5.1}$$

We first look for the optimal loan agreement. The next subsection will discuss its implementation. It is easy to show that it is optimal to have a "cutoff rule" for infusing cash. There exists an optimal threshold ρ^* such that one should continue if and only if

$$\rho \leqslant \rho^*. \tag{5.2}$$

The incentive constraint in the case of continuation is the same as in the absence of a liquidity shock (see Section 3.4):

$$(\Delta p) R_{\mathrm{b}} \geqslant BI. \tag{IC$_{\mathrm{b}}$}$$

The breakeven condition is slightly altered by the presence of liquidity shocks:

$$F(\rho^*)[p_{\mathrm{H}}(RI - R_{\mathrm{b}})] \geqslant I - A + \int_0^{\rho^*} \rho I f(\rho)\, \mathrm{d}\rho. \tag{IR$_{\mathrm{l}}$}$$

That is, the lenders receive a return only if the project is pursued, which has probability $F(\rho^*)$. The left-hand side of (IR$_{\mathrm{l}}$) is the expected pledgeable income. Furthermore, there is a new term, representing the expected outlay on overruns, on the right-hand side. From these two constraints, we deduce the borrowing capacity (or, more precisely, the maximum investment that allows the lenders to break even):

$$I = k(\rho^*) A,$$

where

$$\begin{aligned} k(\rho^*) &= \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho - F(\rho^*)[p_{\mathrm{H}} R - p_{\mathrm{H}} B/\Delta p]} \\ &= \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\, \mathrm{d}\rho - F(\rho^*)\rho_0} \end{aligned} \tag{5.3}$$

exists a $\bar{\rho}$ such that

$$c(\bar{\rho}) \leqslant \hat{c}(\rho^*),$$

which implies that

$$U_b \geqslant \hat{U}_b.$$

In words, corporate risk management lowers the expected unit cost of effective investment and adds value.[31]

Remark (substitutes to corporate hedging: alternative risk transfer). Note that insurance could be provided by means other than hedging on a market. In particular, a bank could offer a *conditional* credit line, such that the maximal amount varies one-to-one (and positively) with ε.[32] Namely, the maximal commitment is equal to $(\rho^* + \varepsilon)I$, and so the firm can withstand liquidity shocks $\rho I \leqslant (\rho^* + \varepsilon)I - \varepsilon I = \rho^* I$. In the absence of transaction costs, conditional credit lines and corporate hedging are perfect substitutes. Such contingent credit lines do exist,[33] but they are less pervasive than corporate hedging. Contingent credit lines may substitute for corporate hedging when either the insurance contract must be tailored to the borrower's specific needs (and so there is no market for the corresponding claims) or when it is difficult to write formal hedging contracts because the underlying shock cannot be well-described *ex ante* or objectively measured *ex post*. The contingent credit line must in the latter circumstances rest on the bank's reputation for abiding by its implicit promises.

In circumstances in which the risk can be insured against in deep markets, corporate hedging is likely to be a lower-transaction-cost alternative; for, and as we will see in future chapters, the credit line is only one of several variables that must be indexed to exogenous shocks such as macroeconomic shocks. (For example, managerial compensation should not depend on shocks over which managers have no control. Hence, bonuses and stock options should be indexed on currency and interest rate fluctuations and on several other exogenous risks. Similarly, the allocation of control rights among claimholders should be indexed on such variables.)

While it may be simpler to have the firm engage in corporate hedging rather than index many contracts and covenants, further study is needed before drawing such a conclusion. As a matter of fact, financing arrangements known under the heading of alternative risk transfer (ART) have developed over the years although they have still much scope for growth. Such products blend elements of corporate finance and insurance. A case in point is catastrophe bonds (cat bonds) such as the ones issued by Vivendi Universal to cover its movie studios in Los Angeles against earthquakes, or the bonds that are contingent on the occurrence of a hurricane.[34]

5.4.2 When Is Incomplete Hedging Optimal? Another Look at the Sensitivity of Investment to Cash Flow

We just obtained a stark result of *full hedging*: any exogenous income fluctuation perturbs optimal liquidity management by making the firm sometimes reinvest when the reinvestment cost is high while it sometimes is unable to reinvest for low reinvestment costs. Even leaving aside the *transaction costs* involved in entering hedging contracts (including those associated with the monitoring of the counterparty's solvency), there are several reasons why firms, or countries for that matter, should not, and actually do not in practice, fully hedge.

function) implies that $H(\bar{\rho}) \leqslant E_\varepsilon[H(\rho^* - \varepsilon)]$, and so $c(\bar{\rho}) \leqslant c(\rho^*)$, as announced.

31. As is the case for the allocation of the initial credit between liquid and illiquid assets (see Section 5.3.3), the borrower's compliance with corporate hedging must be monitored. We invite the reader to check that it may not be in the borrower's best interest to indeed purchase the associated insurance policy once she has obtained the financing for the investment and secured the associated amount of liquidity.

32. Alternatively, the firm could issue debt with interest payments indexed on the shock ε. For example, an oil producer could issue debt whose interest payment increases with the market price of oil.

33. Standby loan commitments are informal arrangements, generally backing the issue of commercial paper by large firms. Under a standby loan agreement, the bank promises to refinance the firm during disruptions in the commercial paper market (Veitch 1992). Incidentally, it is interesting to note that the usage rate on standby commitments is low relative to other categories of loan commitments (Veitch 1992).

In the area of international finance, a number of authors have proposed that reimbursement of sovereign debt be made contingent on observable shocks, such as GDP or exchange rate fluctuations, or (better as this does not give rise to government moral hazard) to world prices of raw materials and other competitive exports of the country.

34. As another example, a few years ago Michelin secured a bank credit line and an insurance facility for five years, contingent on a simultaneous fall in GDP in its various markets and in tyre sales.

(a) *Market power.* Consider the producer of a raw material (copper, oil, etc.) with market power. The market price then depends not only on uncertainty that is exogenous to the firm (e.g., demand shifts), but also on the firm's supply decisions. Thus, suppose for illustrative purposes that there are two dates, 0 and 1 (these two dates are meant to correspond to the risk-management-choice and the risk-income dates of the model). And suppose for simplicity that the firm is a monopolist in the market for the raw material. The monopolist at date 0 sells f units forward at predetermined price p^f. This amounts to writing an insurance contract that pays the firm at date 1 f times the (positive or negative) difference between p^f and the date-1 spot price. Once the monopolist has sold these f units, though, they are no longer hers, and therefore the monopolist has at date 1 decreased incentives to restrain output to keep the spot price up. From the point of view of the monopolist at date 1, output withholding raises the price on her extra production only (her inframarginal units do not include the forward sales). Forward sales overall result in an output that exceeds the monopoly output and therefore reduce revenue.[35]

Example. Suppose that the date-1 spot price is $\tilde{a} - q$, where $\tilde{a}$ is an exogenous demand shock realized at date 1 and q is output, and that the marginal cost is 0. In the absence of forward sales, the monopolist chooses q at date 1 so as to maximize $q(\tilde{a} - q)$, yielding $q = \frac{1}{2}\tilde{a}$ and a revenue that is random at date 0: $r = \frac{1}{4}\tilde{a}^2$. The expected profit is thus $\frac{1}{4}E[\tilde{a}^2]$, where $E[\cdot]$ denotes an expectation with respect to $\tilde{a}$.

Suppose now that the monopolist sells f units at price p^f at date 0. At date 1, the monopolist chooses an extra output q (to be added to the f units that she committed to deliver) so as to maximize $q[\tilde{a} - (q + f)]$, and so $q = \frac{1}{2}(\tilde{a} - f)$.[36] Under rational expectations, the forward price must be equal to the expected spot price:

$$p^f = E[\tilde{a} - (q + f)] = E[\tfrac{1}{2}(\tilde{a} - f)].$$

Total (date-0 plus date-1) profit,

$$\tfrac{1}{4}(E[\tilde{a}^2] - f^2),$$

decreases with f.

More generally, forward sales reduce monopoly power, and so, in the absence of date-1 reinvestment need, it is strictly optimal not to hedge at all ($f = 0$).[37] When one combines the corporate risk management motive of this chapter with the exercise of market power, the optimal degree of hedging is partial hedging.

(b) *Serial correlation of profits.* An important assumption behind the full-hedging result of Section 5.4.1 is that the date-1 profit realization conveys no information about the firm's prospects: it is a transitory shock. Suppose in contrast that a high date-1 profit is good news about date-2 profitability. For example, the price of a crop may reflect permanent shocks such as the reduction of trade barriers, the entry of competing offers, or a change in consumer preferences.

With positive serial correlation of profits, a high current profit is associated with attractive reinvestment opportunities. This suggests that the liquidity available to the borrower at date 1 should covary with the date-1 profit (so, for example, the farmer's debt contract should not be fully indexed to the price of the crop). Things are, however, more complex than this first argument suggests, because better prospects also make it easier for the borrower to return to the capital market at the intermediate stage. The *attractive-reinvestment-opportunities*

35. This reasoning is reminiscent of that underlying the "Coase conjecture," which states that a durable-good monopolist tends to create its own competition and to "flood the market" (see, for example, Tirole 1988, Chapter 1), although the setting is slightly different (the good is here nondurable).

36. We assume that the price is always positive. Otherwise,

$$q = \max\{\tfrac{1}{2}(\tilde{a} - f), 0\};$$

but the gist of the analysis remains the same.

37. This basic insight must be amended a bit in the case of oligopoly. A large literature, starting with Allaz and Vila (1993), has shown that firms that compete *à la* Cournot (in quantities) partially hedge despite the absence of reinvestment need. The intuition is that forward markets induce each oligopolist to try to act as a "Stackelberg leader" and to thereby force its rivals to cut output on the spot market (see, for example, Chao et al. (2005), Creti and Manca (2005), and Willems (2005) for recent contributions to this literature). As usual, this conclusion is reversed if firms compete in prices rather than quantities (see Mahenc and Salanié 2004); under price competition, oligopolists would like to "commit" to set high prices so as to induce others to also set high prices. Buying (i.e., gambling) on the forward market is a commitment for suppliers to set high prices in the spot market.

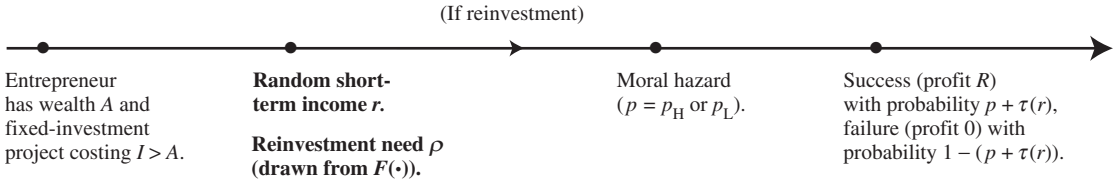


Figure 5.7

effect, however, in general dominates the *easier-refinancing effect*, and so the firm should not be fully insured against exogenous profit shocks, as we now illustrate.

Let us consider the fixed-investment model of Section 5.2, but with two twists:

- The short-term income, r, is random, with mean $\bar{r}$.
- The probability of success in the case of continuation is an increasing function of r,

$$p + \tau(r), \quad \text{with } \tau' > 0,$$

where $p = p_H$ or p_L depending on the entrepreneur's date-1 behavior. (The separable form of the probability-of-success function as usual guarantees that the incentive constraint is invariant.) We assume that the realizations of r and ρ are independent.

These twists are depicted in bold in Figure 5.7.

Let us follow the steps of Section 5.2 and determine the optimal state-contingent cutoff $\rho^*(r)$ (so continuation occurs if and only if $\rho \leqslant \rho^*(r)$). Letting $E[\cdot]$ denote expectations with respect to r, the NPV is

$$U_b = \bar{r} + E[F(\rho^*(r))[p_H + \tau(r)]R] - I - E\left[\int_0^{\rho^*(r)} \rho f(\rho)\, d\rho\right].$$

The investors' breakeven constraint similarly is

$$\bar{r} + E\left[F(\rho^*(r))[p_H + \tau(r)]\left[R - \frac{B}{\Delta p}\right]\right] \geqslant I - A + E\left[\int_0^{\rho^*(r)} \rho f(\rho)\, d\rho\right].$$

Letting μ denote the shadow price of the breakeven constraint (we assume that the constraint is binding, i.e., $\mu > 0$), the first-order condition with respect to $\rho^*(r)$ yields, for each r,

$$\rho^*(r) = \frac{[p_H + \tau(r)][R + \mu(R - B/\Delta p)]}{1 + \mu}.$$

Let us now investigate the implementation of the optimal contract. A *fully indexed debt* can be defined as a date-1 liability $d(r)$ such that

$$d(r) = d_0 + r,$$

for some constant d_0. That is, in the absence of refinancing in the capital market, a fully indexed debt insulates the firm's retained earnings against its cash-flow risk. We, however, want to allow the firm to return to the capital market: insulation of retained earnings against the cash-flow risk does not imply insulation of the reinvestment policy. The amount it can raise in the capital market at date 1,

$$[p_H + \tau(r)]\left(R - \frac{B}{\Delta p}\right),$$

is increasing with the date-1 profit as $\tau' > 0$ (this was referred to earlier as the "easier-refinancing effect"). The optimal policy is implemented when the cutoff is equal to the cash cushion plus the refinancing capacity:

$$\rho^*(r) = [r - d^*(r)] + [p_H + \tau(r)]\left(R - \frac{B}{\Delta p}\right),$$

or

$$d^*(r) = r - \frac{p_H + \tau(r)}{1 + \mu}\left(\frac{B}{\Delta p}\right).$$

In the presence of an agency cost ($B > 0$), the debt is not fully indexed. The easier-refinancing effect is at play, but the existence of a managerial rent puts a limit on what can be achieved by returning to the capital market. Put differently, *the firm should keep some of its cash flow as retained earnings.*

The source of this cash-flow sensitivity of debt is the monotonicity of managerial rents[38] with the

38. Here, this rent is $[p_H + \tau(r)](B/\Delta p)$.

resolution of uncertainty. Thus, the credit rationing problem at the seasoned offering stage is more severe, the more favorable the resolution of uncertainty. While this monotonicity is often a reasonable assumption, one can, of course, envision cases where it does not hold. To check our intuition, Exercise 5.11 considers the case of a permanent price shock P: the date-1 income is Pr (where r is now known and P is a random variable realized at date 1) and the date-2 income in the case of success is PR. The managerial rent in the case of continuation[39] is then insensitive to the state of nature. While the date-1 cash flow affects reinvestment through its informational content, there should not be any cash-flow sensitivity of retained earnings; put differently, debt due at date 1 is perfectly indexed to the output price ($d(P) = Pr - \ell_0$ for some positive ℓ_0).

When, in contrast, a high profit today announces *low* profits tomorrow (negative serial correlation, $\tau' < 0$), the conclusions are reversed. Suppose, for example, that an industry is subject to cycles and furthermore that investments made at the peak (trough) mature at the trough (peak); one possible story is that the other firms in the industry are subject to poor governance and that they invest when they have large cash flows rather than when investments are profitable. How should a (well-governed) firm behave in such an industry? By analogy with the formula above, it should retain less money in net terms when its profit grows.[40]

(c) *Aggregate risk.* Hedging markets often involve economic variables, such as interest rates or exchange rates, that respond to macroeconomic shocks. As is well-known and reflected, for instance, in the capital asset pricing model (CAPM), aggregate risk must be borne by and is optimally shared among economic agents; insuring against it therefore involves a risk premium. Put differently, economic agents cannot insulate themselves from such risks at a "fair price."

We invite the reader to return to the analysis of Section 5.4.1, focusing for simplicity on linear insurance schemes and assuming that eliminating a fraction θ of the income shock (which therefore becomes $(1-\theta)\varepsilon$ in net terms) costs $\sigma\theta$ (proportional to θ). It is easy to see[41] that it is suboptimal to fully hedge; that is, the optimal θ is less than 1. Intuitively, a small risk (θ close to, but smaller than 1) induces only small deviations from the optimal risk management and reinvestment policy, and therefore a second-order NPV loss; in contrast, the cost of this insurance is first order and proportional to θ.

We thus conclude that firms should hedge less against shocks involving larger macroeconomic risk premia.

(d) *Asymmetric information.* Asymmetric information may limit the development of hedging markets. Consider, for example, the potential market for five-year hedges against variations in the overall power prices and in zonal price differences in the U.S. electricity Midwest market. The value of such derivatives depends on very complex predictions of the evolution of supply and demand as well as of likely changes in incentive regulation for both generators and transmission grid owners.

Generators, load-serving entities, and transmission owners, who are keen on hedging their positions, may find few counterparts who have the necessary expertise. And even if some employees of financial institutions do have this expertise, their bosses probably do not and will be reluctant to let them gamble large amounts of money on such long-term derivative markets.

(e) *Incentives.* Finally, borrowers may need to be made somewhat accountable for fluctuations in an exogenous variable, because the quality of their investments depends on how well they predict the future value of this variable. For example, the oil manager of a small oil company has no impact on the oil price; however, the choice of how much to invest in oil rather than in other activities depends on her forecast of the future price of oil. In this case, insulating the borrower from fluctuations in the oil price provides poor incentives for accurate prediction and therefore for efficient investment.

Forecasting future exogenous variables can be modeled in the basic framework as a date-0 moral hazard. The next section studies the implications for liquidity management of such *ex ante* moral hazard.

39. Equal to $p_H(B/\Delta p)$.

40. This policy may be difficult to implement, especially if the firm can hide profits (see Chapter 7).

41. See Holmström and Tirole (2000) for a more rigorous proof.

There, it will be shown that borrowers should not be rewarded for good short-term performance solely through monetary compensation and that liquidity should be sensitive to cash flow. This implies, in particular, that the liquidity of an oil company should not be fully insulated from fluctuations in the stock price even if the company has no market power.

5.5 Endogenous Liquidity Needs, the Sensitivity of Investment to Cash Flow, and the Soft Budget Constraint

5.5.1 Endogenous Liquidity Shocks

Starting with Dewatripont and Maskin (1995), the economics literature has stressed the perverse incentive effects of bailouts and other insurance devices: a state-owned enterprise that knows that it will be bailed out by the government if it loses money has little incentive to reduce its costs or generate revenue.[42] A project manager who knows that the company will be keen on completing the project once large fixed costs have been sunk may "goldplate" the project or spend time on other activities. Hardening the budget constraint may therefore improve incentives.[43]

In the context of corporate financing, liquidity hoarding and credit line commitments become less attractive when liquidity shocks are endogenous, that is, when they depend on the borrower's actions. For incentive purposes, it is not optimal to commit to rescue the borrower often. The borrower has suboptimal incentives to avoid adverse shocks if she knows that she can easily raise cash to cover such shocks. In such circumstances the borrower must be kept "on a short leash." We will discuss how this can be done.

To illustrate in a stark way the point that one may want to commit to a "hard budget constraint," suppose that, after the loan agreement is signed but before the reinvestment need parameter ρ is realized, the borrower can by incurring private effort cost c prevent any cost overrun: $\rho = 0$ with probability 1 (as in Section 3.4). On the other hand, ρ is drawn from distribution $F(\rho)$ (as in this section) if the borrower does not incur this cost. Suppose further that c is small enough that it is optimal to induce the borrower to incur the cost.

Assuming for example that the firm has no date-1 income (and so is cash-poor), the optimal policy then obviously consists in letting the borrower invest I and promising never to plow back any money into the firm. In this case the borrower knows that if she does not spend c, the project will be discontinued with probability 1 (provided that the cumulative distribution F has no atom at 0). This threat obviously keeps her on her toes.

The crux of the matter is then, How can we make this hard budget constraint credible? For, we have seen that, in the case of "reasonable" overrun ($\rho \leqslant \rho_0$), the lenders have an *ex post* incentive to renege on their promise not to rescue the firm. Anticipating this, the borrower may not bother to incur cost c to prevent overruns.

5.5.1.1 A Broader Perspective

When is the firm's budget constraint likely to be soft? The basic idea of long-term financing is, as we have seen in Sections 5.2 and 5.3, that the intermediate stage (date-1) exhibits rationing of credit for reinvestment and so it is optimal for the firm to secure *ex ante* (at date 0) more liquidity than it will obtain by going to the capital market at the intermediate stage. Thus the problem is not that the capital market is too soft but rather that it is too tough at the intermediate stage. Hence, the soft-budget-constraint problem does not arise.

This need not be so, however, when information accrues at date 1 that sheds light on some activity subject to earlier (date-0) moral hazard.[44] It is then optimal to commit at date 0 to punish the entrepreneur if the information "signals" that the borrower has not acted in the lenders' interest.

42. See Kornai (1980) for a study of the soft budget constraint in centrally planned economies and its macroeconomic consequences.

43. Hardening the budget constraint may, however, induce short-termism, that is, a managerial focus on immediate performance, to the detriment of long-term goals, as was demonstrated by von Thadden (1995). See Chapter 7 for a study of short-termism.

44. Or to adverse selection for that matter. For example, if information accrues at date 1 that the entrepreneur is likely to be a bad borrower, notwithstanding claims to the contrary at the contracting stage, it is in general optimal to commit at date 0 to punish the firm for such bad news at date 1, in order to screen borrowers more efficiently. See Chapter 6 for a treatment of adverse selection.

Recall also the debate between Fazzari et al. (1988) and Kaplan and Zingales (1997) as to whether firms with a weak balance sheet exhibit a higher sensitivity of investment to cash flow. We took a first look at this prediction in Section 3.2.7 by interpreting "cash flow" as "net worth" and observed that the theory makes no clear prediction in this regard. In that section, though, we argued that this first look has drawbacks and that firms are better viewed as ongoing entities.

The relationship

$$\rho^*(r) - E[\rho^*(r)] = \lambda \ell(r)$$

indicates that (re)investment should indeed be sensitive to cash flow: continuation or investment (in the reinterpretation in which retentions are used to finance growth prospects) are part of an optimal carrot-and-stick scheme designed to encourage the production of cash flow.[53]

The issue of whether the sensitivity of investment to cash flow increases with the intensity of financial constraints is more complex. In the case of small date-0 moral hazard (implying $\Delta(r) \equiv 0$), and letting $\hat{\rho} \equiv E[\rho^*(r)]$, the constraint ($IC_b'$) can be rewritten as

$$E_r[F(\hat{\rho} + \lambda \ell(r))\ell(r)] = \frac{B_0}{\rho_1 - \rho_0}.$$

For a uniform distribution ($F(\rho) = f \cdot \rho$) and using the fact that the expectation of the likelihood ratio is equal to 0, we obtain

$$\lambda E_r[\ell^2(r)] = \frac{B_0}{f(\rho_1 - \rho_0)} = \text{constant}.$$

The financial constraint impacts only the average liquidity in that, as earlier, a tighter financial constraint in general results in a shorter maturity structure:[54]

$$\rho^*(r \mid A) = \hat{\rho}(A) + \lambda \ell(r).$$

Thus, *for a uniform distribution, the sensitivity of investment to cash flow is independent of the financial constraint.*[55] More generally, with nonuniform distributions, the sensitivity parameter λ may increase or decrease with A. We thus conclude that no strong prediction emerges as to the relationship between financial constraint and sensitivity of investment to cash flow.

5.6 Free Cash Flow

As we discussed in the introduction to this chapter, the free-cash-flow problem faced by firms with excess liquidity is the mirror image of the liquidity shortage problem faced by cash-poor ones. While the latter must contract on the provision of liquidity beyond the level provided *ex post* by the capital market, the former must design a mechanism that forces them to pay out excess cash in the future.

We first review the relationship between the liquidity shortage and the free-cash-flow problems. The problem of preventing inefficient liquidation of cash-poor firms becomes one of preventing inefficient continuation of the cash-rich firm. This results in a theory of claim maturity. The optimal contract takes the form of a mandatory payment to claimholders at date 1. As in Section 5.2.2, this payment, which can be interpreted either as a dividend as in Easterbrook (1984)[56] or as short-term debt as in Jensen (1986), forces the borrower to pay out the excess cash and prevents her from wasting it on suboptimal reinvestments.

Section 5.6.2 goes beyond this reinterpretation of the liquidity shortage model by considering more complex settings in which a fixed payment is not optimal. As has been emphasized in the literature, rough instruments such as short-term debt then simultaneously allow some undesirable reinvestments and prevent some desirable ones. As we explain, optimal contracting requires the firm to use market information more fully in order to properly manage the firm's liquidity.

53. As discussed in Chapter 2, $\rho^*(r)$ alternatively should increase with r even in the absence of date-0 moral hazard, if the first- and second-period revenues are correlated. A simple way to introduce this learning effect in our model would be to assume that the date-2 probability of success is $p + \tau(r)$, where (i) $p = p_L$ or p_H depends, as usual, on the entrepreneur's date-1 behavior, and (ii) τ is increasing in r (see (b) in Section 5.4.2).

54. (IR_I), in the case of a uniform distribution and normalizing $f = 1$, can be rewritten as

$$\hat{\rho}\rho_0 - \tfrac{1}{2}\hat{\rho}^2 - \lambda^2 E[\tfrac{1}{2}\ell^2(r)] = I - A - \bar{r}.$$

Because λ is independent of A and $\hat{\rho} > \rho_0$, $\hat{\rho}$ increases with A.

55. The constant-returns-to-scale model, as usual, is not appropriate to study the impact of the intensity of financial constraints on the sensitivity of investment to cash flow, since all firms are scaled-up or scaled-down versions of each other (Program NSBC depends only on A/I). But suppose that I is fixed in Program NSBC (more generally, returns could be decreasing).

56. An early paper on dividends with a similar idea is Rozeff (1982).

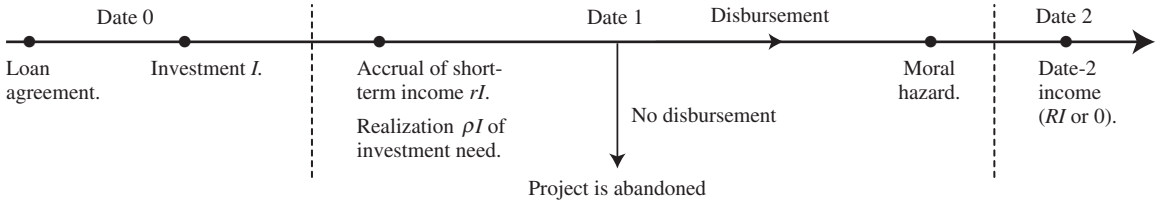


Figure 5.10

5.6.1 Optimal Claim Maturity

Let us return to the continuous-investment, continuous-shock version of Section 5.3.2, but with a short-term income (the analysis is not really new and is therefore only sketched): see Figure 5.10. Because the short-term income rI is fully pledgeable to the lenders, everything is *as if* the unit investment cost were equal to $1 - r$ instead of 1. The lenders' breakeven condition, that is, the equality between expected revenue and expected investment cost, becomes

$$rI + F(\rho^*)\rho_0 I = I - A + \left[\int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right] I$$

and so

$$k(\rho^*) = \frac{1}{1 + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho - [r + F(\rho^*)\rho_0]}. \quad (5.3'')$$

The margin (expected profit of the firm per unit of investment) becomes

$$m(\rho^*) = [r + F(\rho^*)\rho_1] - \left[1 + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho\right]. \quad (5.4'')$$

And thus the borrower's (gross) utility becomes

$$U_\mathrm{b} = m(\rho^*)k(\rho^*)A = \frac{\rho_1 - c(\rho^*)}{c(\rho^*) - \rho_0}A,$$

where the expected unit cost of effective investment, $c(\rho^*)$, is given by

$$c(\rho^*) = \frac{1 - r + \int_0^{\rho^*} \rho f(\rho)\,\mathrm{d}\rho}{F(\rho^*)}. \quad (5.5'')$$

So the optimal threshold is given by

$$\int_0^{\rho^*} F(\rho)\,\mathrm{d}\rho = 1 - r \quad (5.6'')$$

and the borrower's utility by

$$U_\mathrm{b} = \frac{\rho_1 - \rho^*}{\rho^* - \rho_0}A. \quad (5.7'')$$

It is important to note that *the short-term income, even though it is deterministic and fully pledgeable, is not equivalent to an increase in the borrower's cash on hand A*. Such an increase in equity would result in a larger investment (as is the case here), but not in a modification of the continuation rule. By contrast, condition (5.6″) shows that the larger the short-term profit, the lower the optimal threshold ρ^*. To understand this point, recall the tradeoff between increasing borrowing capacity (by choosing ρ^* close to ρ_0) and increasing the probability of continuation (by choosing ρ^* close to ρ_1). The short-term revenue (like a salvage value) makes it worth sacrificing continuation more in order to boost borrowing capacity. Lastly, note that the distinction between a short-term revenue and a salvage value is that the salvage value is obtained only if the investment is liquidated at date 1. And so the net expected date-2 profit and date-1 pledgeable income are $\rho_1 - L$ and $\rho_0 - L$ in the case of a salvage value, and ρ_1 and ρ_0 in the case of a short-term income. This explains the difference between, say, (5.7′) and (5.7″).

(a) *Liquidity management.* Let us now turn to the implementation of the optimum and thus to the claim maturity. To this purpose we make the following assumption.[57]

Free-cash-flow assumption: $r > \rho^*$. Under the free-cash-flow assumption, and given that the entrepreneur cannot steal the intermediate income, the entrepreneur would reinvest excessively if she were not asked to pay out money to investors at date 1. Namely, she would reinvest as long as $\rho \leqslant r$.

To obtain the optimal amount of reinvestment, an amount $P_1 \equiv (r - \rho^*)I$ must be pumped out of the firm, and the entrepreneur must be denied the right to dilute initial investors.

57. Of course, it must also be the case that $\rho^* > \rho_0$ (otherwise, the borrower's borrowing capacity and utility would be infinite in this constant-returns-to-scale model). Because $\mathrm{d}\rho^*/\mathrm{d}r < -1$, we must thus also assume that r is not "too large."

Remark (salvage value). The analysis is again extended straightforwardly to allow for a salvage value LI for the assets if the project is discontinued at date 1. The threshold ρ^* is then given by

$$\int_0^{\rho^*} F(\rho)\,\mathrm{d}\rho = 1 - r - L.$$

We thus conclude that the short-term payment $P_1 = (r - \rho^*)I$ grows faster than the salvage value.

5.6.2 Liquidity Management in More General Settings

The previous section considered a somewhat special setting, in which short-term debt suffices to fine-tune the firm's cash at date 1. As one might imagine, a fixed payment at date 1 in general is unlikely to be quite the right way to manage a cash-rich firm's liquidity (neither is a fixed credit line for a cash-poor firm). More instruments are needed in order to obtain the optimal state-contingent reinvestment policy. A sizeable literature has developed that shows that with rough instruments such as short-term debt there is in general a tradeoff between allowing more undesirable reinvestments and preventing more desirable ones (see, for example, Harris and Raviv 1990; Hart and Moore 1995; Stulz 1991).[58] The literature has not yet, to the best of my knowledge, come to grips with a general theory of liquidity management. Although we will not provide such a theory, we can make a number of observations relative to it.

Investors' date-1 control of liquidity is unlikely to be optimal. One might think that date-1 control by investors provides the flexibility required when a fixed payment (or a fixed credit line) does not properly adjust the firm's liquidity. We have seen, however, that investors tend to liquidate excessively (to reinvest too little), and so investors' control is unlikely to be optimal.

Make full use of market information. Consider a general environment in which a number of variables besides the liquidity shock are random and are realized and publicly observed at date 1: the first-period income r, the salvage value L, the second-period expected payoff in the case of continuation ρ_1 and the date-1 pledgeable income in the case of continuation ρ_0. Suppose in a first step that these variables are verifiable by a court of law. Then the optimal contract should specify a state-contingent threshold $\rho^*(r, L, \rho_1, \rho_0)$ beyond which reinvestment does not take place. This state-contingent threshold is straightforwardly computed by generalizing the previous analysis to random payoff values (see below for an example of such a computation).

At date 1, though, only the first-period income r is directly verifiable. The implementation of the optimal state-contingent rule requires extracting the values of L, ρ_1, and ρ_0 from the capital market. The date-1 values of the securities provide such information; in this respect, we should note that a diversity of tradable securities creates more market valuations and may be able to "span" a larger state space. But reading from market valuations is not the only way to extract information about the state of nature. For example, the acceptance of an exchange offer by a secured creditor with unpaid short-term debt (that is, an offer of securities or cash in exchange for debt forgiveness) reveals information about the salvage value L of the assets that are collateralized. Similarly, the renegotiation of existing claims embodies available information at date 1.[59]

Again, our aim here is not to develop a general theory of liquidity management, but rather to point out that optimal liquidity management should make use of the wealth of information held by the capital market about current and future asset values. We now illustrate this point through an example.

5.6.2.1 An Illustration: *Ex Ante* Uncertainty about the Second-Period Income

Let us assume that there is *ex ante* uncertainty not only about the liquidity shock ρ but also about the

58. In Harris and Raviv and Stulz, short-term debt reduces free cash flow. Hart and Moore allow a more complex management of liquidity (they allow the amount of cash used at date 1 to be contingent on the date-2 revenue, which is deterministic at date 1 in their model). They do not, however, allow the firm's liquidity to be fully contingent on the market's information about variables that are realized at date 1 and which could be obtained from the value of securities or the money raised in a security issuance.

59. David (2001), for example, argues that the renegotiation of putable securities enables the payment to their holders to be contingent on the state of nature.

second-period income in the case of success (see Exercise 5.8 for a different illustration of the use of market valuations for liquidity management). More precisely, suppose that the second-period income is equal to RI with probability $\underline{\alpha}$ and to $(R+\Delta R)I$ (with $\Delta R > 0$) with probability $\bar{\alpha} = 1 - \underline{\alpha}$. (All of our results generalize to a continuum of possible values for the income in the case of success.) The second-period income in the case of failure is always equal to 0. So, in terms of our general modeling, there is uncertainty of magnitude $p_H \Delta R$ with regards to both ρ_0 and ρ_1. For notational simplicity, we set $L = 0$ (no salvage value).

One can show that if all variables were verifiable at date 1, the optimal liquidity management would specify two thresholds, $\underline{\rho}^*$ when the second-period income is RI and $\bar{\rho}^*$ when the second-period income is $(R + \Delta R)I$, where[60]

$$\bar{\rho}^* = \underline{\rho}^* + p_H \Delta R.$$

As one would expect, the optimal threshold moves one-to-one with the realized increment in expected second-period income and pledgeable income.

It is clear that *short-term debt is no longer sophisticated enough to provide the firm with the appropriate amount of liquidity*. Assuming away any right for the entrepreneur to dilute initial investors, a fixed payment P_1 defines a threshold,

$$\rho^* \equiv r - P_1,$$

that is independent of the news about date-2 income.

It is also easy to illustrate in this model a tradeoff that has been highlighted repeatedly in the literature. Suppose one constrains oneself to the use of short-term debt and that the firm is not allowed to conduct a seasoned offering at date 1. The optimal level of short-term debt defines a threshold equal to the value given in (5.6′),[61] and satisfying

$$\bar{\rho}^* > \rho^* > \underline{\rho}^*.$$

So, under the restriction to liquidity management through short-term debt, the contract must trade off insufficient reinvestment in the state in which prospects are good and excessive reinvestment in the state in which the prospects are mediocre. And, indeed, at the constrained optimum, there is excessive reinvestment when $\rho \in (\underline{\rho}^*, \rho^*]$ and the second-period per-unit income in the case of success is R and insufficient reinvestment when $\rho \in (\rho^*, \bar{\rho}^*]$ and the second-period per-unit income in the case of success is $(R + \Delta R)$.

A similar point can, of course, be made for a random date-1 income, as a fixed P_1 does not pump the proper amount of money out of the firm as long as either r or ρ^* is random. This tradeoff thus suggests why (nonindexed) *debt is a more appropriate instrument for firms with safe cash flows* (regulated public utilities, banks, firms in mature industries).[62]

To let the reinvestment policy respond to future prospects, it is necessary to use market information about these prospects. There are several ways of doing so. Here is a simple way of relying efficiently on market information in the context of an unknown payoff in the case of success: force the entrepreneur to pay out

$$P_1 = [r - (\underline{\rho}^* - \rho_0)]I$$

(if P_1 is positive; otherwise contract at date 0 for a credit line at level $-P_1$); and give the entrepreneur the right to dilute at date 1 existing securities in order to withstand a liquidity shock. Because the pledgeable income is equal to $\rho_0 I$ in the mediocre state and $(\rho_0 + p_H \Delta R)I$ in the good state,

60. We leave it to the reader to check that

$$k(\underline{\rho}^*, \bar{\rho}^*) = \frac{1}{(1-r) - \underline{\alpha}[F(\underline{\rho}^*)\rho_0 - \int_0^{\underline{\rho}^*} \rho f(\rho)\,\mathrm{d}\rho] - \bar{\alpha}[F(\bar{\rho}^*)(\rho_0 + p_H\Delta R) - \int_0^{\bar{\rho}^*} \rho f(\rho)\,\mathrm{d}\rho]}$$

and

$$m(\underline{\rho}^*, \bar{\rho}^*) = \underline{\alpha}\left[F(\underline{\rho}^*)\rho_1 - \int_0^{\underline{\rho}^*} \rho f(\rho)\,\mathrm{d}\rho\right] + \bar{\alpha}\left[F(\bar{\rho}^*)(\rho_1 + p_H\Delta R) - \int_0^{\bar{\rho}^*} \rho f(\rho)\,\mathrm{d}\rho\right] - (1-r).$$

61. The reader will check that

$$U_b = \frac{(\rho_1 + \bar{\alpha} p_H \Delta R) - \rho^*}{\rho^* - (\rho_0 + \bar{\alpha} p_H \Delta R)} A,$$

where ρ^* is given by (5.6′).

62. Jensen and Meckling (1976) argue on different grounds that firms with safe cash flows should have more debt. They are interested in the conflict of interests between shareholders and debtholders, and observe that high debt levels may induce shareholders to pursue highly risky strategies if the riskiness of income can be easily manipulated. Note that the definitions of "safe cash flows" are not quite the same in both arguments. We use "safe" in the sense of "nonrisky" while Jensen and Meckling emphasize the absence of moral hazard in the choice of riskiness.

the entrepreneur is able to withstand shocks up to

$$rI - [r - (\underline{\rho}^* - \rho_0)]I + \rho_0 I = \underline{\rho}^* I$$

in the mediocre state and

$$rI - [r - (\underline{\rho}^* - \rho_0)]I + (\rho_0 + p_H \Delta R)I = \bar{\rho}^* I$$

in the good state. We have thus verified that the use of market information about date-2 income allows the implementation of the optimal state-contingent reinvestment policy.

It is clear that more sophisticated mechanisms are required to fine-tune the firm's liquidity when there is also uncertainty about the first-period income,[63] the salvage value, and the entrepreneur's minimum stake (which defines $\rho_1 - \rho_0$). But the general message is clear: market mechanisms can supply the information that is required to implement an optimal liquidity management policy.

5.7 Exercises

Exercise 5.1 (long-term contract and loan commitment). Consider the two-project, two-period version of the fixed-investment model of Section 3.2 and a unit discount factor. Assume, say, that the borrower initially has no equity ($A = 0$). Show the following.

(i) If $p_H(p_H R - I) + (p_H R - I - p_H B/\Delta p) \geqslant 0$, then the optimal long-term contract specifies a loan commitment in which the second-period project is financed at least if the first-period project is successful. Show that if $p_H(p_H R - I) + (p_H R - I - p_H B/\Delta p) > 0$, then the optimal long-term contract specifies that the second-period project is implemented with probability 1 in the case of first-period success, and with probability $\xi \in (0, 1)$ in the case of failure.

(ii) In question (i), look at how ξ varies with various parameters.

(iii) Is the contract "renegotiation proof," that is, given the first-period outcome, would the parties want to modify the contract to their mutual advantage?

(iv) Investigate whether the long-term contract outcome can be implemented through a sequence of short-term contracts where the first-period contract specifies that the borrower receives $\bar{A} = I - p_H(R - B/\Delta p)$ with probability 1 in the case of success and with probability ξ in the case of failure.

Exercise 5.2 (credit rationing, predation, and liquidity shocks). (i) Consider the fixed-investment model. An entrepreneur has cash A and can invest $I_1 > A$ in a project. The project's payoff is R_1 in the case of success and 0 otherwise. The entrepreneur can work, in which case her private benefit is 0 and the probability of success is p_H, or shirk, in which case her private benefit is B_1 and the probability of success p_L. The project has positive NPV ($p_H R_1 > I_1$), but will not be financed if the contract induces the entrepreneur to shirk. The (expected) rate of return demanded by investors is 0.

What is the threshold value of A such that the project is financed?

In the following, let

$$\rho_0^1 \equiv p_H\left(R_1 - \frac{B_1}{\Delta p}\right).$$

The next three questions add a prior period, period 0, in which the entrepreneur's equity A is determined. The discount factor between dates 0 and 1 is equal to 1.

(ii) In this question, the entrepreneur's date-1 (entire) equity is determined by her date-0 profit. This profit can take one of two values, a or A, such that

$$a < I_1 - \rho_0^1 < A.$$

At date 0, the entrepreneur faces a competitor in the product market. The competitor can "prey" or "not prey." The entrepreneur's date-0 profit is a in the case of predation and A in the absence of predation. Preying reduces the competitor's profit at date 0, but by an amount smaller than the competitor's date-1 gain from the entrepreneur's date-1 project not being funded.

- What happens if the entrepreneur waits until date 1 to go to the capital market?
- Can the entrepreneur avoid this outcome? You may want to think about a credit line from a

63. If this first-period income rI is random but exogenous (that is, not affected by moral-hazard or adverse-selection considerations), it suffices to distribute it, so as not to create a spurious dependence of the reinvestment policy on the particular realization of date-1 income. The optimal policy is clearly more complex if moral-hazard or adverse-selection considerations imply that the entrepreneur should be rewarded for high date-1 income by receiving more liquidity.

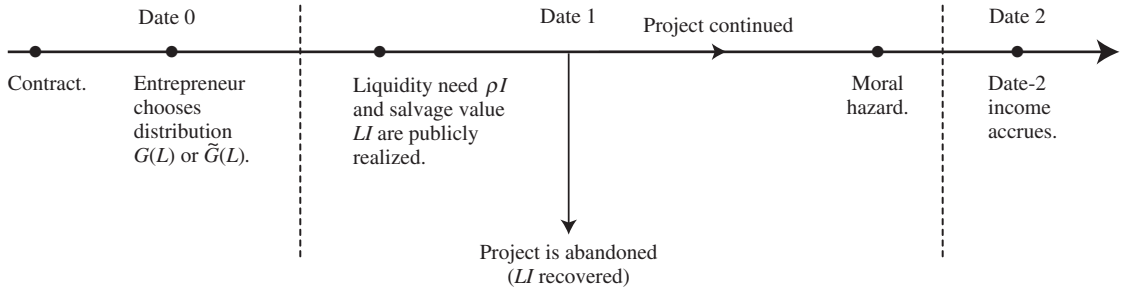


Figure 5.11

bank. Would such a credit line be credible, that is, would it be renegotiated to the mutual advantage of the entrepreneur and his investors at the end of date 0?

(iii) Forget about the competitor, but keep the assumption that the entrepreneur's date-0 profit can take the same two values, a and A. We now introduce a date-0 moral-hazard problem on the entrepreneur's side.

Assume that the entrepreneur's date-0 production involves an investment cost I_0 and that the entrepreneur initially has no cash. The entrepreneur can work or shirk at date 0. Working yields no private benefit and probability of profit A equal to $q_{\rm H}$ (and probability $1 - q_{\rm H}$ of obtaining profit a). Shirking yields private benefit B_0 to the entrepreneur, but reduces the probability of profit A to $q_{\rm L} = q_{\rm H} - \Delta q$ ($0 < q_{\rm L} < q_{\rm H} < 1$). Assume that

$$I_1 + I_0 - (q_{\rm L}A + (1 - q_{\rm L})a) > \rho_0^1.$$

- Interpret this condition.

Consider the following class of long-term contracts between the entrepreneur and investors. "The date-1 project is financed with probability 1 if the date-0 profit is A and with probability $x < 1$ if this profit is a. The entrepreneur receives $R_{\rm b} = B_1/\Delta p$ if the date-1 project is financed *and* succeeds, and 0 otherwise." Assume that such contracts are not renegotiated.

- What is the optimal probability x^*? (Assume that $(\Delta q)p_{\rm H}B_1 \geqslant (\Delta p)B_0$.)
- Assuming that $\rho_0^1 > I_1$, is the previous contract robust to (a mutually advantageous) renegotiation?

(iv) Show that the entrepreneur cannot raise sufficient funds at date 0 if renegotiation at the end of date 0 cannot be prevented, if $\rho_0^1 > I_1$, and if

$$I_0 + I_1 - (q_{\rm H}A + (1 - q_{\rm H})a) > \rho_0^1 - \left(\frac{q_{\rm H}B_0}{\Delta q}\right).$$

Exercise 5.3 (asset maintenance and the soft budget constraint). Consider the variable-investment framework of Section 5.3.2, except that the date-0 moral hazard affects the per-unit salvage value L. Date-1 income is now equal to a constant (0, say). Assets are resold at price LI in the case of date-1 liquidation. The distribution of L on $[0, \bar{L}]$ is $G(L)$, with density $g(L)$, if the borrower works at date 0, and $\tilde{G}(L)$, with density $\tilde{g}(L)$, if the borrower shirks at date 0. We assume the monotone likelihood ratio property:

$$\frac{g(L)}{\tilde{g}(L)} \text{ is increasing in } L.$$

The borrower enjoys date-0 private benefit $B_0 I$ if she shirks, and 0 if she shirks. The timing is summarized in Figure 5.11.

As usual, let $\rho_1 \equiv p_{\rm H}R$ and $\rho_0 = p_{\rm H}(R - B/\Delta p)$. And let

$$\ell(L) \equiv \frac{g(L) - \tilde{g}(L)}{g(L)}.$$

(i) Determine the optimal contract $\{\rho^*(L), \Delta(L)\}$ (where $\rho^*(L)$ and $\Delta(L)$ are the state-contingent threshold and extra rent (see Section 5.5.2)) in the absence of the soft budget constraint (that is, the commitment to the contract is credible). Show that

- $\rho^*(L) = -L + (\rho_1 + \mu\rho_0 + \nu(\rho_1 - \rho_0)\ell(L))/(1 + \mu)$ for some positive μ and ν;
- $\Delta(L) = 0$ as long as $\rho^*(L) \leqslant \rho_1 - L$;
- conclude as to when rewards take the form of an increased likelihood of continuation or cash (or both).

(ii) When would the investors want to rescue the firm at date 1 if it has insufficient liquidity? Draw $\rho^*(L)$ and use a diagram to provide a heuristic description of the soft-budget-constraint problem. Show that the soft budget constraint arises for $L \leqslant L_0$ for some $L_0 \geqslant 0$.

Exercise 5.4 (long-term prospects and the soft budget constraint). Perform the same analysis as in Exercise 5.3, with the difference that the date-0 choice of the entrepreneur does not affect the salvage value, which is always equal to 0. Rather, the date-0 moral hazard refers to the choice of the distribution of the second-period income in the case of continuation. This income is $R_L \geqslant 0$ or $R_H = R_L + R$, where $R \geqslant 0$ is a constant. The distribution of R_L, $G(R_L)$, or $\tilde{G}(R_L)$ is determined at date 0. Assume that $g(R_L)/\tilde{g}(R_L)$ is increasing in R_L. As usual, let p_H and p_L denote the probabilities of R_H when the entrepreneur works or shirks *ex post*. And let $\rho_1 \equiv p_H R$ and $\rho_0 = p_H(R - B/\Delta p)$. Assume that R_L is publicly revealed at date 1 before the continuation decision. Solve for the optimal state-contingent policy in the absence of the soft-budget-constraint problem. Show that the soft-budget-constraint problem arises (if it arises at all) under some threshold value of R_L.

Exercise 5.5 (liquidity needs and pricing of liquid assets). Consider the liquidity-needs model with a *fixed investment* and *two possible liquidity shocks*. The borrower has cash A and wants to finance a fixed-size investment $I > A$ at date 0. At date 1, a cash infusion equal to ρ is needed in order for the project to continue. If ρ is not invested at date 1, the project stops and yields nothing. If ρ is invested, the borrower chooses between working (no private benefit, probability of success p_H) and shirking (private benefit B, probability of success $p_L = p_H - \Delta p$). The project then yields, at date 2, R in the case of success and 0 in the case of failure.

The liquidity shock is equal to ρ_L with probability $(1 - \lambda)$ and to ρ_H with probability λ, where

$$\rho_L < \rho_0 < \rho_H < \rho_1,$$

where $\rho_1 \equiv p_H R$ and $\rho_0 \equiv p_H(R - B/\Delta p)$. Assume further that

$$\rho_0 - \rho_L > I - A. \tag{1}$$

There is a single liquid asset, Treasury bonds. A Treasury bond yields 1 unit of income for certain at date 1 (and none at dates 0 and 2). It is sold at date 0 at price $q \geqslant 1$. (The investors' rate of time preference is equal to 0.)

(i) Suppose that the firm has the choice between buying enough Treasury bonds to withstand the high liquidity shock and buying none. Show that it chooses to hoard liquidity if

$$(q-1)(\rho_H - \rho_0) \leqslant (1-\lambda)(\rho_0 - \rho_L) - \lambda(\rho_H - \rho_0) - I + A \tag{2}$$

and

$$(q-1)(\rho_H - \rho_0) \leqslant \lambda(\rho_1 - \rho_H). \tag{3}$$

(ii) Suppose that the economy is composed of a continuum, with mass 1, of identical firms with characteristics as described above. The liquidity shocks of the firms are perfectly correlated. There are T Treasury bonds in the economy, with $T < \rho_H - \rho_0$. Show that when λ is small, the liquidity premium $(q - 1)$ commanded by Treasury bonds is proportional to the probability of a high liquidity shock. (Hint: show that either (2) or (3) must be binding, and use (1) to conclude that (3) is binding.)

(iii) Suppose that, in the economy considered in the previous subquestion, the government issues at date 0 not only the T Treasury bonds, but also a security that yields at date 1 a payoff equal to 1 in the good state (the firms experience liquidity shock ρ_L) and 0 in the bad state (the firms experience liquidity shock ρ_H). What is the equilibrium date-0 price q' of this new asset? (Prices of the Treasury bonds and of this new asset are market clearing prices.)

Exercise 5.6 (continuous entrepreneurial effort; liquidity needs). (i) An entrepreneur with initial cash A and protected by limited liability wants to invest in a fixed-size project with investment cost $I > A$. After the investment is made, the entrepreneur chooses the probability p of success ($0 \leqslant p \leqslant 1$); the disutility of effort is $g(p) = \frac{1}{2}p^2$. (The entrepreneur enjoys no private benefit in this model.) In question (i) only, the profit is $R = 2\sqrt{I - A}$ in the case of success and 0 in the case of failure. (We assume that $R < 1$ to avoid considering probabilities of success exceeding 1. R takes an arbitrary value in question (ii).) As usual, the uninformed investors demand

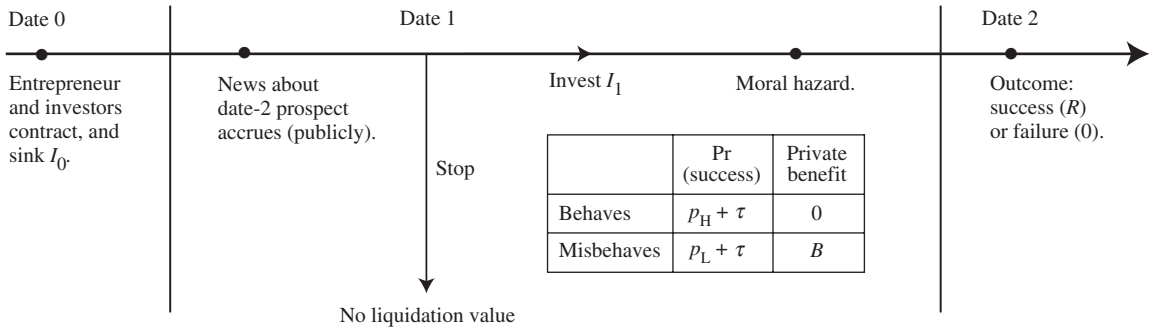


Figure 5.12

an expected rate of interest equal to 0 and everyone is risk neutral. Let R_b denote the entrepreneur's reward in the case of success.

Solve for the optimal contract (R_b). Show that

$$R_b = \tfrac{1}{2}R.$$

(ii) Now introduce an intermediate liquidity shock ρ (for a now arbitrary level A of cash on hand). The cumulative distribution of ρ is $F(\rho)$ on $[0, \infty)$ and the density $f(\rho)$. The effort decision is made *after* the value of ρ is realized and, of course, *conditional* on the choice of continuing (incurring reinvestment cost ρ). *Suppose* that the entrepreneur's stake in the case of continuation (R_b) is independent of ρ. Write the investors' breakeven condition. Write the optimization program yielding (R_b, ρ^*), where ρ^* is the cutoff liquidity shock.

Exercise 5.7 (decreasing returns to scale). Extend the treatment of Section 5.6.1 to the case of decreasing returns to scale: the payoff in the case of continuation and success is $R(I)$, with $R(0) = 0$, $R' > 0$, $R'' < 0$, $R'(0) = \infty$, and $R'(\infty) = 0$. The rest is unchanged (the short-term income is rI, the reinvestment need is ρI, and the private benefit is BI).

(i) What are the first-order conditions yielding the optimal investment level I and cutoff ρ^*?

(ii) Assuming that $r > \rho^*$, and that $(R(I)/I - R'(I))$ is increasing in I (a condition satisfied, for example, by $R(I)$ quadratic), derive the impact of the strength of the balance sheet as measured, say, by A on debt maturity.

Exercise 5.8 (multistage investment with interim accrual of information about prospects). In this chapter we have focused mostly on the case of shocks about the reinvestment need (cost overruns, say). Consider, instead, the case of news about the final profitability. In the two-outcome framework, news can accrue about either the probability of success or the payoff in the case of success. We consider both, in sequence. The investment is a multistage one: let

$$I = I_0 + I_1,$$

where I_0 is the date-0 investment and I_1 is the date-1 reinvestment. In contrast with I_0, I_1 is not incurred if the firm decides to stop. The timing is as in Figure 5.12.

As usual, the entrepreneur has initial wealth A, is risk neutral, and protected by limited liability. Investors are risk neutral. The discount rate is equal to 0. If reinvestment cost I_1 is sunk at date 1, then the firm can continue. Misbehavior reduces the probability of success by Δp, but yields private benefit B to the entrepreneur.

Assume

$$B < (\Delta p)R.$$

As announced, we consider two variants.

(a) *News about the probability of success.* R is known at date 0, but the probability of success is $p_H + \tau$ in the case of good behavior and $p_L + \tau$ in the case of misbehavior, where τ is publicly learned at the beginning of date 1. The random variable τ is distributed according to the distribution function $F(\tau)$ with density $f(\tau)$ on $[\underline{\tau}, \bar{\tau}] = [-p_L, 1 - p_H]$ (to keep probabilities in the interval $[0, 1]$). Let τ^e denote the expectation of τ.

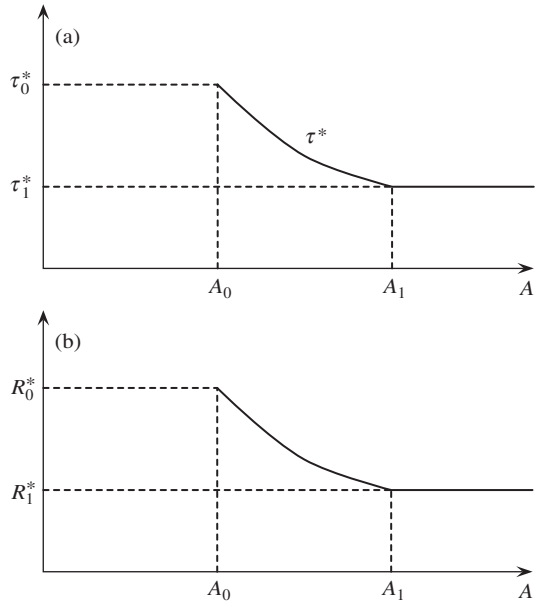


Figure 5.13

(b) *News about the payoff in the case of success.* The probabilities of success are known: p_H and p_L (normalize: $\tau = 0$). By contrast, the profit R in the case of success is drawn from distribution $G(R)$ with density $g(R)$ on $(0, \infty)$. (The profit in the case of failure is always equal to 0.)

(i) For each variant, show that there exist two thresholds, A_0 and A_1, $A_0 < A_1$, such that the first best prevails for $A \geqslant A_1$ and financing is secured if and only if $A \geqslant A_0$. Show that the continuation rules take the form of cutoffs, as described in Figure 5.13.

Determine τ_0^*, τ_1^*, R_0^*, R_1^*.

(ii) For each variant, assume that $A = A_0$. Let $y \equiv (p_H + \tau)R$ denote the expected income, and $\mathcal{R}(y)$ denote the entrepreneur's rent in the case of continuation. Show that (above the threshold y^*)

- $0 < \mathcal{R}'(y) < 1$ in variant (a);
- $\mathcal{R}$ is constant in variant (b).

Exercise 5.9 (the priority game: uncoordinated lending leads to a short-term bias). This chapter, like Chapters 3 and 4, has assumed that the firm's balance sheet is transparent. In particular, each investor has perfect knowledge of loans made by other lenders and of the firm's obligations to them.

This exercise argues that uncoordinated lending leads to financing that is too oriented to the short term. In a nutshell, lenders, by cashing out early, exert a negative externality on other lenders. Because this externality is not internalized, the resulting financial structure contains too much short-term debt.

We consider a three-period model: $t = 0, 1, 2$. The entrepreneur has no cash ($A = 0$), is risk neutral, and is protected by limited liability. At date 0, a fixed investment I is made. The project yields a known return $r > 0$ at date 1, and an uncertain return (R or 0) at date 2. Because the point is quite general and does not require credit constraints, we assume away moral hazard; or, equivalently, the private benefit from misbehaving is 0. The probability of a date 2 success is

$$p + \tau(I_1),$$

where I_1 is the date-1 deepening investment, equal to r minus the level of short-term debt repaid to lenders and the date-1 payment to the entrepreneur (the firm does not return to the capital market at date 1), and τ is an increasing and concave function (with $\tau'(0) = \infty$). Assume that $\tau'(r)R < 1$.

We assume that the entrepreneur cannot engage in "fraud," that is, cannot fail to honor the short-term debt and, if the project succeeds at date 2, the long-term debt. By contrast, obligations to lenders, and in particular I_1, cannot be verified as the firm's balance sheet is opaque.

(i) Derive the first-best investment I_1^*. Show how this allocation can be implemented by a mixture of short- and long-term debt (note that in this model without moral hazard the structure of compensation for the entrepreneur exhibits a degree of indeterminacy).

(ii) Assume that $r - I_1^* < I$ (creditors must hold long-term debt). Suppose next that financing is not transparent. Start from the first-best solution, with a large number (a continuum of mass 1) of lenders, with the representative lender owning short-term claim r_1 and contingent long-term claim R_1 on the firm.

Show that the entrepreneur has an incentive to secretly collude with any lender to increase the latter's short-term claim in exchange for a smaller long-term claim.

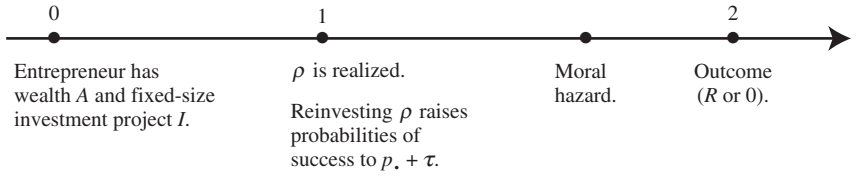


Figure 5.14

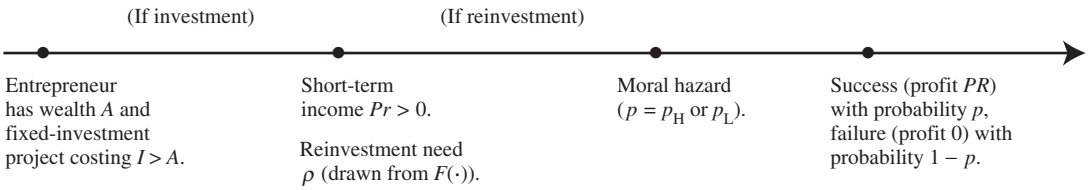


Figure 5.15

Given the constraint that financing is provided by many lenders and that the latter do not observe each other's contracts, is the indeterminacy mentioned in question (i) resolved?

Exercise 5.10 (liquidity and deepening investment). (i) Consider the fixed-investment model. The entrepreneur has cash A and can invest $I > A$ in a project. The project's return in the case of success (respectively, failure) is R (respectively, 0). The probability of success is p_H if the entrepreneur behaves (she then gets no private benefit) and $p_L = p_H - \Delta p$ if she misbehaves (in which case she gets private benefit B). In this subquestion and in the subsequent extension, one will assume that the project is viable only if the incentive scheme induces the entrepreneur to behave. The entrepreneur and the capital market are risk neutral; the entrepreneur is protected by limited liability; and the market rate of interest is equal to 0.

Let

$$\rho_1 \equiv p_H R \quad \text{and} \quad \rho_0 \equiv p_H[R - B/\Delta p],$$

and assume $\rho_1 > I > \rho_0$.

What is the necessary and sufficient condition for the project to be financed?

(ii) Now add an intermediate stage, in which there is an option to make a deepening investment. This investment increases the probability of success to $p_H + \tau$ (in the case of good behavior) and $p_L + \tau$ (in the case of misbehavior).

If the deepening investment is not made, the probabilities of success remain p_H and p_L, respectively. This deepening investment costs ρ, where ρ is unknown *ex ante* and distributed according to distribution $F(\rho)$ and density $f(\rho)$ on $[0, \infty)$. The timing is summarized in Figure 5.14.

Let $\mu \equiv \tau / p_H$, $\hat{\rho}_1 \equiv \mu\rho_1$, and $\hat{\rho}_0 \equiv \mu\rho_0$.

Write the incentive compatibility constraint and (for a given cutoff ρ^*) the investors' breakeven condition.

(iii) What is the optimal cutoff ρ^*? (Hint: consider three cases, depending on whether

$$\rho_0[1 + \mu F(\hat{\rho}_k)] \lessgtr I - A + \int_0^{\hat{\rho}_k} \rho f(\rho)\, d\rho,$$

with $k = 0, 1$.)

(iv) Should the firm content itself with returning to the capital market at date 1 in order to finance the deepening investment (if any)?

Exercise 5.11 (should debt contracts be indexed to output prices?). This exercise returns to optimal corporate risk management when profits are positively serially correlated (see Section 5.4.2). The source of serial correlation is now a permanent shift in the market price of output, as summarized in Figure 5.15. The model is the fixed-investment model, except that the date-1 and date-2 incomes depend on an exogenous market price P, with mean $\bar{P}$, that is realized at date 1. The realizations of P and ρ are independent.

The rest of the model is otherwise the same as in Section 5.2. Following the steps of Section 5.4.2:

(i) Determine the optimal reinvestment policy $\rho^*(P)$.

(ii) Show that, accounting for seasoned offerings, the optimal debt is *fully indexed debt*:

$$d(P) = Pr - \ell_0,$$

where ℓ_0 is a positive constant.

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